Formula 1 Podium Prediction Companion

Milestone 1 — Analytical Report

Course: DMET904
Dataset: Formula 1 World Championship (1950–2024)
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Kaggle Notebook Link:
https://www.kaggle.com/code/darsh2984/acl-project-darsh-sara-farah-aesha-merged
GitHub Repository: https://github.com/Darsh2984/ACL-Sem-09-Project-
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1. Introduction

Formula 1 is a motorsport championship in which race outcomes depend on several interacting factors, including driver performance, constructor competitiveness, starting grid position, reliability, and race conditions. Predicting race outcomes is challenging because these factors change across seasons and different technical eras.

This project develops a machine-learning pipeline to predict whether a Formula 1 driver will finish a race in one of the top three positions, commonly known as the podium.

The prediction is formulated as a binary classification problem:

- **Podium (1):** The driver finishes in first, second, or third place.
- **Non-podium (0):** The driver does not finish in the top three.

Predictions are made after qualifying and before the race begins. Consequently, only information available before the race is permitted as a predictor. Race results, pit stops, finishing positions, and updated post-race standings are excluded to prevent data leakage.

The project covers the Formula 1 World Championship from 1950 through 2024 and uses historical driver and constructor performance alongside starting-grid information.

Three machine-learning approaches are investigated:

1. Logistic Regression
2. Random Forest Classifier
3. Shallow Neural Network (Multilayer Perceptron)

The models are compared using ROC-AUC, Precision-Recall AUC, and F1-score. Additional experiments investigate feature-group contributions, preprocessing choices, and model interpretability.

2. Dataset Description

2.1 Data Source

The project uses the Formula 1 World Championship dataset available through Kaggle, based on historical Ergast Formula 1 records.

The original dataset is relational and contains 14 tables describing races, circuits, drivers, constructors, results, qualifying sessions, lap times, pit stops, standings, and other championship information.

These tables are joined using appropriate identifiers to construct an analytical dataset suitable for podium prediction.

The unit of analysis is one driver entering one race. Accordingly, the combination of `raceId` and `driverId` must be unique.

2.2 Final Modeling Dataset

After the previously completed cleaning and duplicate-handling process, the modeling dataset contains:

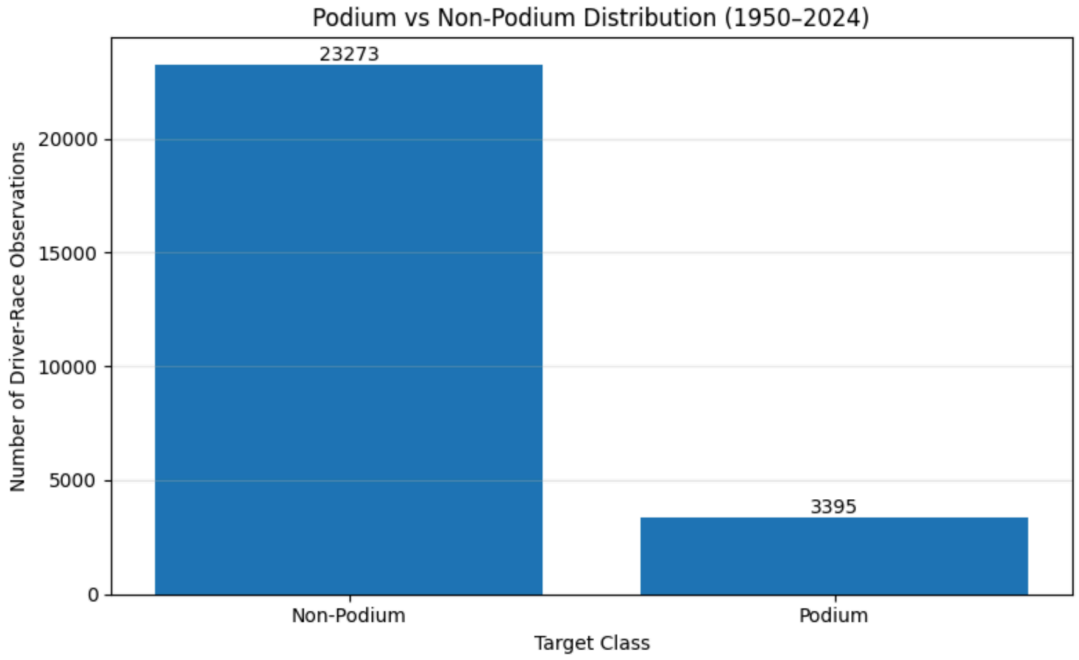
Property	Value
Historical period	1950–2024
Driver-race observations	26,668
Races	1,125
Drivers	861

Constructors	210
Podium observations	3,395
Non-podium observations	23,273
Podium percentage	12.73%
Selected predictors	9

The class distribution indicates that podium finishes represent a minority of the observations.

This imbalance is expected because only a small number of drivers can finish in the top three positions in each race.

Consequently, accuracy alone is insufficient for evaluating the predictive models. Particular attention is given to Precision-Recall AUC, precision, recall, and F1-score.



3. Data Cleaning and Auditing

3.1 Data Quality Requirements

Before constructing the predictive dataset, the source tables require cleaning and validation.

The cleaning process addresses missing-value sentinels, data types, duplicate driver-race records, and relational integrity.

The original source tables are preserved, while cleaning operations are performed on working copies.

3.2 Missing Values and Semantic Types

The dataset uses special sentinel values, including `\N`, to represent unavailable information.

Such values must be converted into proper missing values before numerical calculations, feature engineering, or model training.

Date columns must be represented using date-compatible types, while numerical measurements and time durations must be converted into suitable numeric representations where required.

Missing-value sentinels such as `\N` were treated as missing data rather than valid numerical values. Numerical predictors were subsequently handled through training-fitted median imputation. Date fields were converted to date-compatible representations where needed for chronological processing. The original source tables were preserved, and cleaned working copies were used for subsequent analysis.

3.3 Duplicate Driver-Race Records

A fundamental requirement is that each driver appears only once per race.

Historical Formula 1 records contain unusual cases, particularly shared drives during the 1950s, which can produce multiple records for the same driver in a single race.

Historical duplicate driver-race entries were resolved using a deterministic selection rule. For each `(raceId, driverId)` combination, records with a positive starting grid position were prioritized because they are more likely to represent an actual starting entry. When multiple candidate records remained, the record with the smallest `positionOrder` was retained.

The cleaned dataset was then deduplicated using `(raceId, driverId)` as the composite key, retaining one row per driver per race.

This procedure removed 91 duplicate records and produced a modeling table containing 26,668 driver-race observations, with no remaining duplicate driver-race pairs.

The rule provides a consistent representation of historical shared-drive and irregular-entry cases while maintaining the required prediction grain.

```
Original results rows: 26759
Cleaned results rows: 26668
Rows removed: 91
Remaining duplicated (raceId, driverId) pairs: 0
```

3.4 Relational Integrity

Relational integrity was verified before constructing the final modeling dataset to ensure that source tables could be joined reliably without introducing invalid references or ambiguous identifiers.

Two types of integrity checks were performed: primary-key validation and foreign-key validation.

Primary-Key Validation

Primary-key checks were conducted on seven source tables: circuits, constructors, drivers, races, results, qualifying, and status. Each table was examined for missing primary-key values and duplicate primary-key values.

Table	Records	Missing Primary Keys	Duplicate Primary Keys
circuits	77	0	0
constructors	212	0	0
drivers	861	0	0
races	1,125	0	0

results	26,759	0	0
qualifying	10,494	0	0
status	139	0	0

All seven tables passed the primary-key checks, with no missing or duplicate primary-key values.

Foreign-Key Validation

Foreign-key validation was performed on the results table to verify that its referenced race, driver, constructor, and status identifiers existed in their corresponding parent tables.

Foreign-Key Relationship	Unique Referenced IDs	Missing References
results.raceId → races.raceId	1,125	0
results.driverId → drivers.driverId	861	0
results.constructorId → constructors.constructorId	211	0
results.statusId → status.statusId	137	0

All four tested foreign-key relationships passed, with zero missing references.

Although the constructors table contains 212 records, only 211 constructor identifiers appear in the results table. Similarly, 137 of the 139 status identifiers are referenced. These differences indicate unused parent records rather than invalid foreign-key relationships.

Conclusion

The integrity audit confirmed that all seven tested primary keys were unique and non-null and that all four tested foreign-key relationships contained no invalid references.

These results support the reliability of the relational joins used during data engineering and feature construction.

The original results table contained 26,759 records. Following the separate driver-race deduplication process, 91 duplicate driver-race entries were removed, producing the final analytical dataset of 26,668 observations.

This distinction is important because the original `resultId` primary key can be unique even when multiple records share the same `(raceId, driverId)` combination.

```
Checking primary key: circuits.circuitId
Rows: 77
Missing key values: 0
Duplicate key values: 0
Unique key values: 77

Checking primary key: constructors.constructorId
Rows: 212
Missing key values: 0
Duplicate key values: 0
Unique key values: 212

Checking primary key: drivers.driverId
Rows: 861
Missing key values: 0
Duplicate key values: 0
Unique key values: 861

Checking primary key: races.raceId
Rows: 1125
Missing key values: 0
Duplicate key values: 0
Unique key values: 1125

Checking primary key: results.resultId
Rows: 26759
Missing key values: 0
Duplicate key values: 0
Unique key values: 26759

Checking primary key: qualifying.qualifyId
Rows: 10494
Missing key values: 0
Duplicate key values: 0
Unique key values: 10494

Checking primary key: status.statusId
Rows: 139
Missing key values: 0
Duplicate key values: 0
Unique key values: 139
```

```
Checking foreign key: results.raceId -> races.raceId
Unique child values: 1125
Unique parent values: 1125
Missing references: 0

Checking foreign key: results.driverId -> drivers.driverId
Unique child values: 861
Unique parent values: 861
Missing references: 0

Checking foreign key: results.constructorId -> constructors.constructorId
Unique child values: 211
Unique parent values: 212
Missing references: 0

Checking foreign key: results.statusId -> status.statusId
Unique child values: 137
Unique parent values: 139
Missing references: 0
```

3.5 Cleaning Impact

The impact of data cleaning was evaluated by comparing the original results table with the cleaned analytical dataset.

Metric	Value
Original results records	26,759
Cleaned driver-race records	26,668
Duplicate records removed	91
Remaining duplicate `(raceId, driverId)` pairs	0

The original results table contained 26,759 records. Following the deterministic duplicate-resolution procedure described in Section 3.3, 91 excess driver-race records were removed, reducing the dataset to 26,668 observations.

This represents a reduction of approximately 0.34% of the original results records.

The cleaned dataset contains no remaining duplicate `(raceId, driverId)` combinations, satisfying the requirement that each driver-race pair be represented by exactly one observation.

Although the number of removed records is relatively small, this cleaning step is important because duplicate driver-race entries could distort podium-rate calculations, historical feature engineering, and model evaluation.

The before-and-after comparison confirms that the duplicate-handling procedure successfully enforced the intended analytical grain while retaining approximately 99.66% of the original results records.

```
Original results rows: 26759
Cleaned results rows: 26668
Rows removed: 91
Remaining duplicated (raceId, driverId) pairs: 0
```

4. Exploratory Data Analysis

Exploratory data analysis is used to investigate historical race patterns, class imbalance, and relationships between pre-race predictors and podium finishes.

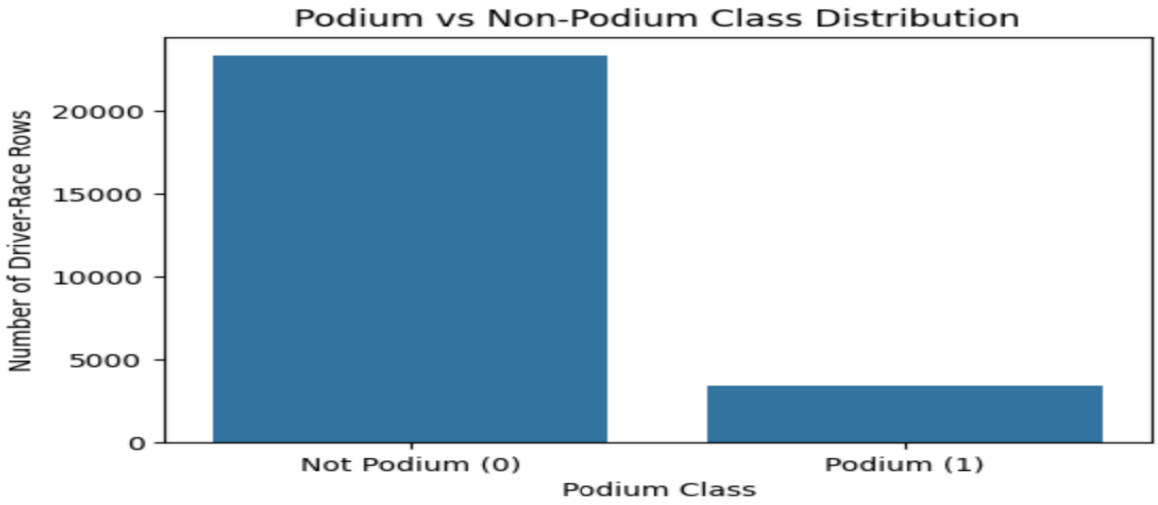
The analysis also supports feature selection by identifying variables that have predictive relevance while remaining available before the race begins.

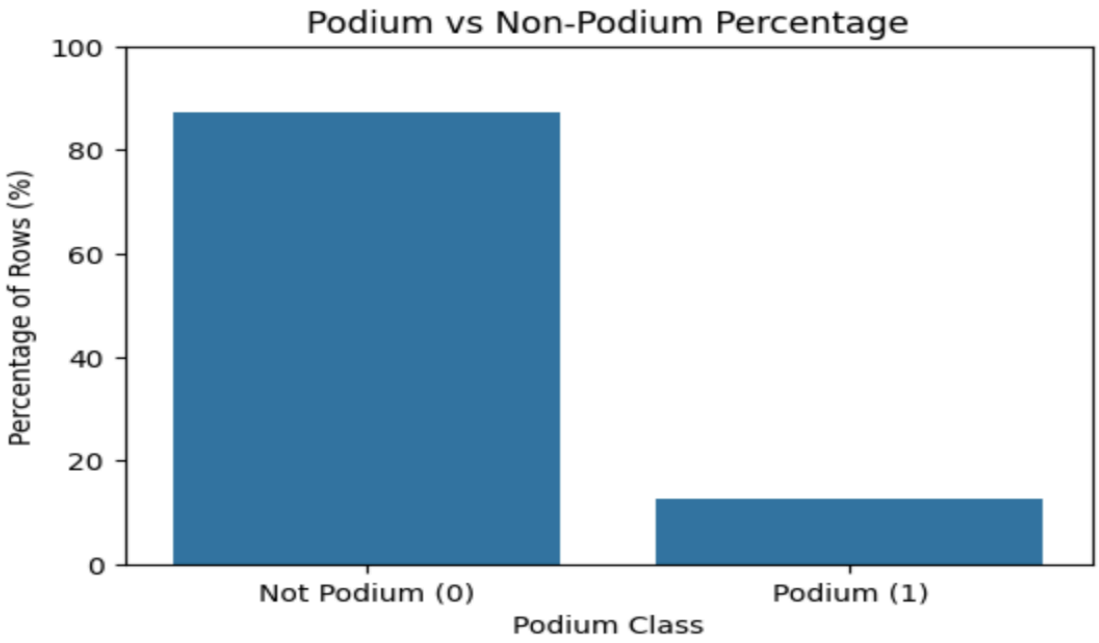
4.1 Podium Class Distribution

The final modeling dataset contains 3,395 podium observations, corresponding to approximately 12.73% of all driver-race entries.

The remaining 87.27% correspond to non-podium finishes.

This imbalance motivates the use of Precision-Recall AUC and F1-score as important evaluation metrics.

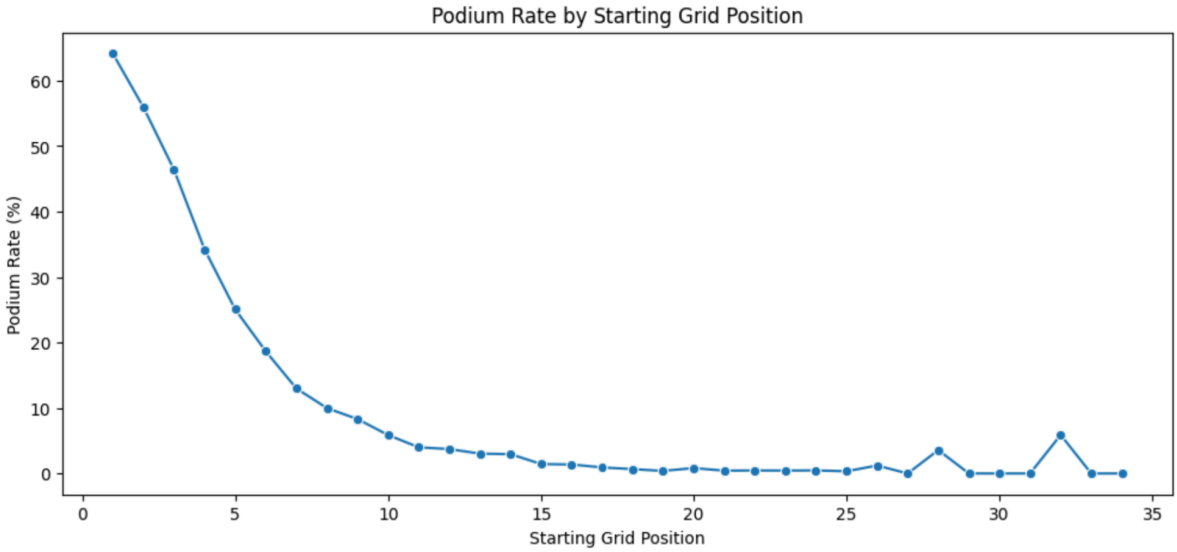




4.2 Starting Grid Position and Podium Probability

Starting grid position is an important pre-race predictor because it reflects a driver's competitive position before the race begins. To investigate its relationship with podium outcomes, the historical podium rate was calculated for each positive starting-grid position.

Figure 4: Podium Finish Rate by Starting Grid Position



The analysis reveals a strong negative relationship between starting-grid position and podium probability. Drivers starting from pole position (P1) achieved a podium rate of approximately 64%, compared with approximately 56% for P2 and 46% for P3. The podium rate declined to approximately 25% at P5 and 6% at P10.

Beyond approximately P15, podium finishes became relatively rare, with most starting positions exhibiting rates close to zero. Occasional increases among drivers starting near the back of the grid were observed, but these should be interpreted cautiously because the number of observations at such positions may be small.

These findings demonstrate that drivers starting closer to the front are substantially more likely to finish on the podium. Consequently, starting-grid position was retained as a central predictive feature. The observed relationship also supports the inclusion of the `front_row_start` indicator, which identifies drivers starting in P1 or P2.

Although the association is strong, it should not be interpreted as proof that grid position alone causes podium finishes, since driver ability, constructor competitiveness, and qualifying performance are also related to starting position.

4.3 Historical Changes Across Seasons

Formula 1 has undergone substantial changes in championship structure, participation, regulations, and competition since 1950. To examine how the volume of historical observations varies across seasons, the number of driver-race records was calculated for each championship year.

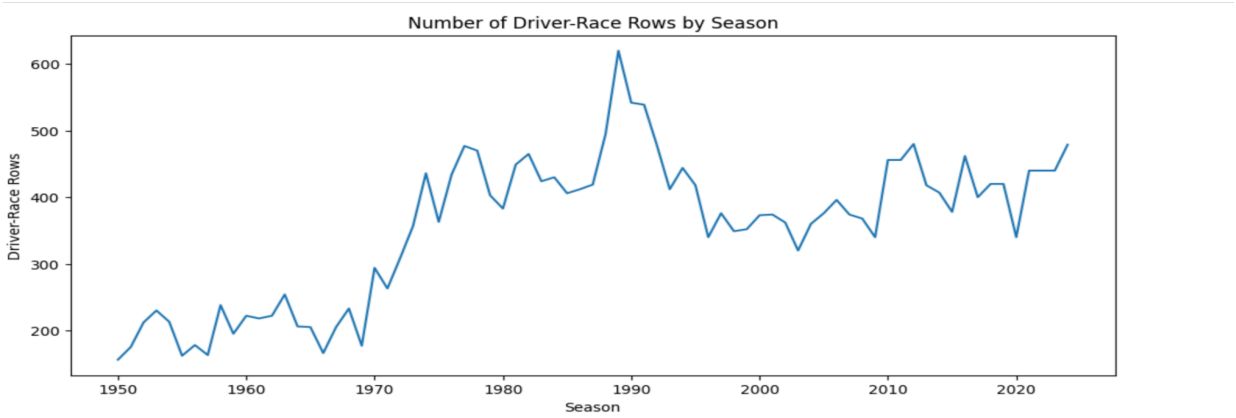


Figure 5: Number of Driver-Race Observations by Season (1950–2024)

The visualization demonstrates considerable variation in the number of driver-race observations across the historical dataset. During the 1950s and 1960s, annual counts were generally between approximately 150 and 250 observations. The number of observations increased during the 1970s and 1980s, reaching a peak above 600 in the late 1980s.

Following this peak, the number of observations declined and fluctuated throughout the 1990s and 2000s. In more recent seasons, the dataset generally contains approximately 350–480 driver-race observations per year.

These variations may reflect changes in the number of championship races, participating drivers, and constructor entries, as well as differences in historical data coverage. The graph describes observation counts rather than podium rates and therefore cannot directly establish changes in the probability of achieving a podium finish.

The observed variation is relevant to predictive modeling because historical seasons contribute different numbers of observations to the training data. Combined with changes in sporting regulations and competitive conditions, this reinforces the importance of chronological splitting when evaluating performance on later seasons.

5. Data-Engineering Analysis

This section addresses the three required analytical questions using joined relational data, appropriate aggregation levels, and supporting visualizations.

5.1 Front-Row Starts and Podium Conversion

Research question: Which circuits convert a front-row start into a podium most frequently, and does the ranking change after the 2014 hybrid era?

The analysis joined historical race results with race and circuit information. A front-row starter was defined as a driver whose actual starting grid position was 1 or 2. The conversion rate for each circuit was calculated as the number of podium finishes by front-row starters divided by the total number of front-row starts at that circuit.

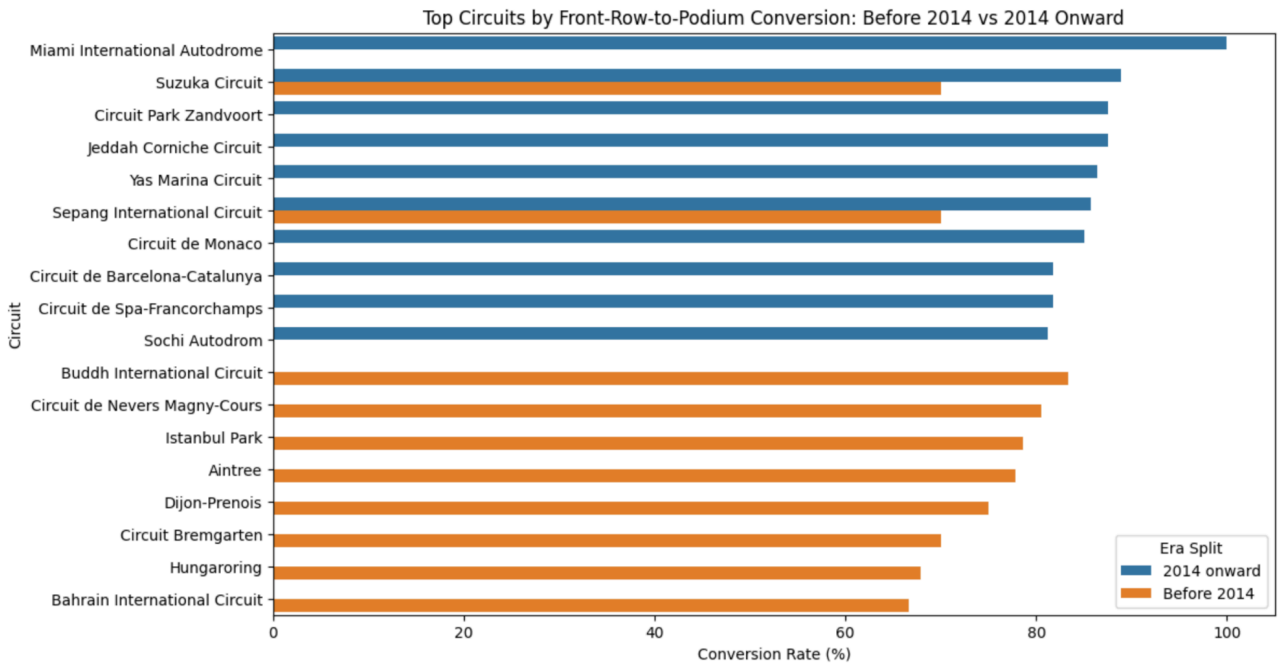
Across the dataset, approximately 60.05% of front-row starts resulted in podium finishes, compared with approximately 8.38% of non-front-row starts.

The circuit rankings changed between the pre-2014 period and the 2014-onward hybrid era. Before 2014, highly ranked circuits included Buddh International Circuit, Circuit de Nevers Magny-Cours, Istanbul Park, Aintree, and Dijon-Prenois. In the 2014-onward period, highly ranked circuits included Miami International Autodrome, Suzuka Circuit, Circuit Park Zandvoort, Jeddah Corniche Circuit, and Yas Marina Circuit.

To avoid rankings dominated by very small samples, the analysis applied a minimum of 10 front-row starts for the overall circuit ranking and six front-row starts per circuit-era combination for the historical comparison.

Actual starting grid position was used rather than qualifying position because grid penalties and other race-weekend decisions can change a driver's starting position after qualifying.

Conclusion: Front-row starting position is strongly associated with podium finishes, but the circuits with the highest observed conversion rates differ across historical eras. These descriptive differences do not establish that circuit characteristics or regulation changes caused the observed outcomes.



5.2 Home-Country Advantage

Research question: Do drivers finish on the podium more frequently when racing in their home country after controlling for starting grid position?

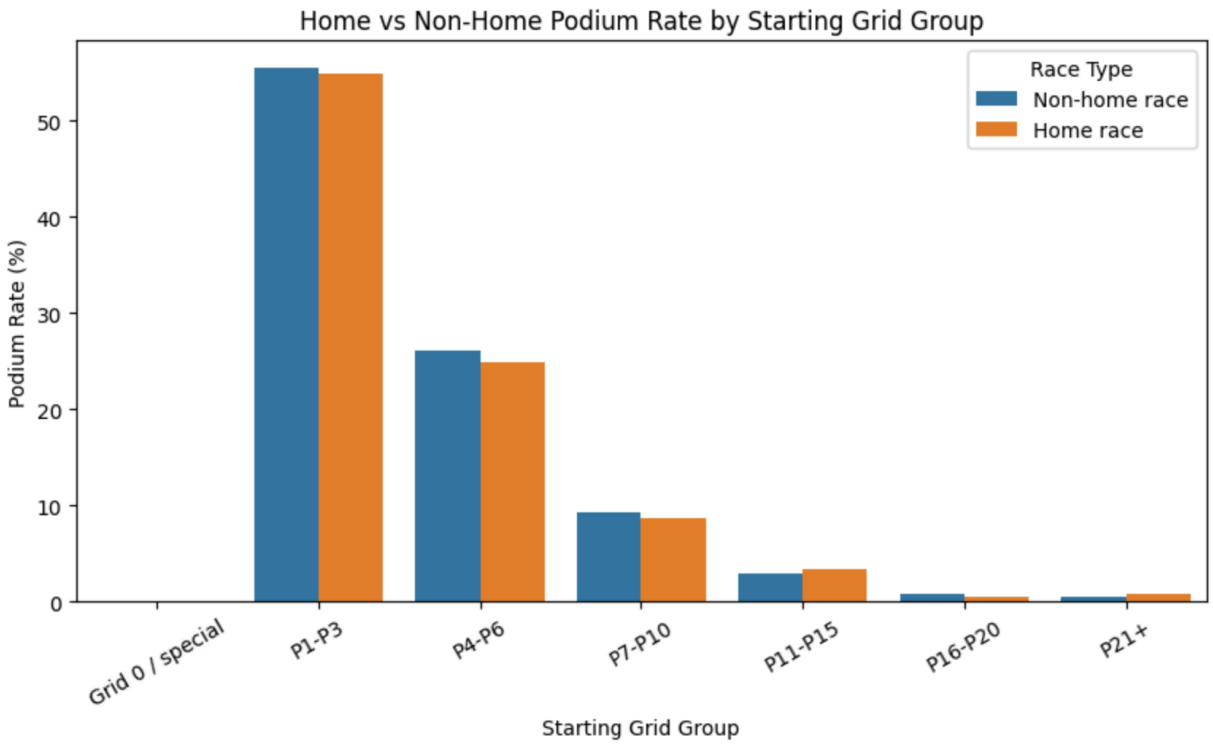
Driver nationality was mapped to a standardized country value and compared with the country of the race circuit. This produced a binary `home_race` indicator. The analysis used driver-race observations and grouped starting positions into comparable categories: Grid 0/special, P1–P3, P4–P6, P7–P10, P11–P15, P16–P20, and P21+.

Across all observations, the podium rate was approximately 10.65% for home-race entries, compared with 12.95% for non-home-race entries. However, this overall comparison does not account for starting position.

After stratifying by grid position, the analysis found no consistent home-race advantage. Among drivers starting P1–P3, the podium rates were approximately 54.92% for home-race entries and 55.54% for non-home-race entries. Home-race podium rates were slightly lower in the P4–P6 and P7–P10 groups, but slightly higher in the P11–P15 and P21+ groups.

Conclusion: The observed data do not demonstrate a consistent home-country podium advantage after comparing drivers within similar starting-grid groups. Starting position appears to be a more influential predictive factor.

This is a descriptive stratified comparison, not a causal estimate. Nationality-to-country mapping and small podium counts in lower-grid groups are limitations.



5.3 Constructor Mechanical Reliability

Research question: Which constructors have the highest mechanical-retirement rates in 2014–2021 compared with 2022–2024?

The analysis joined the results, races, constructors, and status tables to identify constructor race starts and their recorded outcomes. Mechanical retirements were identified using status descriptions associated with component or technical failures, including engine, gearbox, brakes, hydraulics, electrical systems, suspension, cooling, and power-unit failures.

Crashes, collisions, driver-related failures, disqualifications, withdrawals, and classified finishes were excluded from the mechanical-retirement category.

For each constructor and period, the mechanical-retirement rate was calculated as:

Mechanical-retirement rate = Mechanical retirements / Race starts × 100

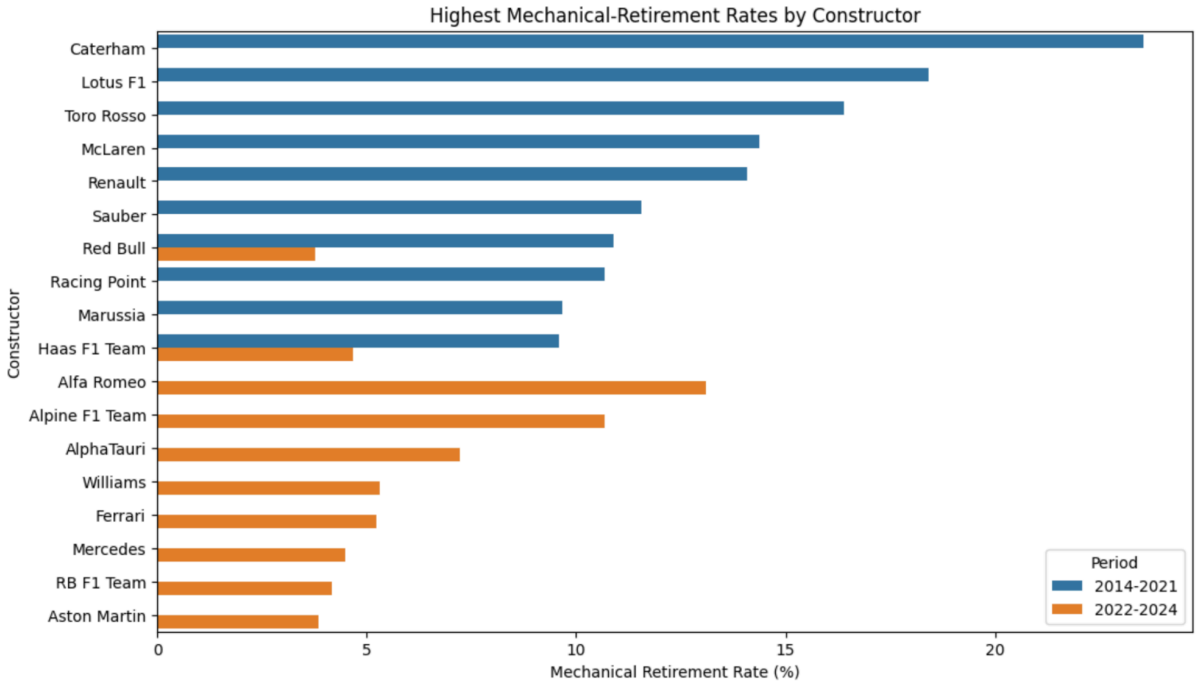
The analysis retained constructor-period combinations with at least 20 race starts.

During 2014–2021, the constructors with the highest observed mechanical-retirement rates included Caterham, Lotus F1, Toro Rosso, McLaren, and Renault. Caterham ranked highest, recording eight mechanical retirements from 34 starts, equivalent to 23.53%.

During 2022–2024, the highest-ranked constructors included Alfa Romeo, Alpine F1 Team, AlphaTauri, Ferrari, and Williams. Alfa Romeo ranked highest, with ten mechanical retirements from 88 starts, equivalent to 11.36%.

Conclusion: The highest observed constructor mechanical-retirement rate was lower in the 2022–2024 period than in 2014–2021. However, this comparison of the highest-ranked constructors does not by itself establish that overall Formula 1 mechanical reliability improved.

The findings depend on the available status descriptions and the manual classification of technical failures. Ambiguous statuses and differences in constructor participation are limitations.



For the mechanical-retirement analysis, race starts were approximated by records with a positive starting-grid position. This definition may exclude legitimate pit-lane or other nonstandard starters recorded with grid position zero. Consequently, the reported retirement rates should be interpreted under this operational definition of a race start.

6. Feature Selection and Engineering

6.1 Prediction Cutoff and Leakage Prevention

All model inputs must be available after qualifying and before the race begins.

Current-race finishing positions, points, status, pit stops, and post-race standings are not valid predictors because their values are only known after the race.

Historical performance features are calculated using previous races only.

This ensures that a driver's current race outcome does not influence the predictors used to forecast that same race.

6.2 Selected Features

Nine features are used across all three models.

Feature	Description
`grid`	Driver's starting grid position
`front_row_start`	Whether the driver starts in the first two grid positions
`grid_zero_flag`	Indicator for a grid value of zero
`driver_previous_races`	Number of previous race entries for the driver
`driver_past_podium_rate`	Historical podium proportion for the driver
`driver_recent_podium_rate_5`	Podium rate across the driver's previous five races
`constructor_previous_races`	Number of previous races represented in constructor history
`constructor_past_podium_rate`	Historical podium proportion for the constructor

`constructor_recent_podium_rate_5`	Constructor podium performance over the previous five races

The historical features are designed to represent accumulated experience and recent performance.

Driver history is calculated using past-only observations. Constructor history is calculated at the constructor-race level before shifting to prevent information from another driver in the same race from leaking into the current prediction.

Empirical Justification of the Selected Features

Feature selection was supported by exploratory analysis of starting-grid position, driver historical performance, constructor historical performance, and correlations among candidate predictors.

The training-set correlation analysis identified a strong positive correlation between driver recent podium rate and constructor recent podium rate ($r = 0.781$). Driver historical podium rate and driver recent podium rate were also positively correlated ($r = 0.706$), as were constructor historical podium rate and constructor recent podium rate ($r = 0.695$).

Starting grid position showed negative correlations with historical and recent performance features, consistent with successful drivers and constructors tending to start closer to the front.

Although several features contain overlapping information, they represent different aspects of competitiveness: current-race starting position, cumulative historical performance, and recent form.

The nine selected predictors were retained for the initial modeling experiments, and their contributions were subsequently assessed using feature-group ablation and model explainability methods.

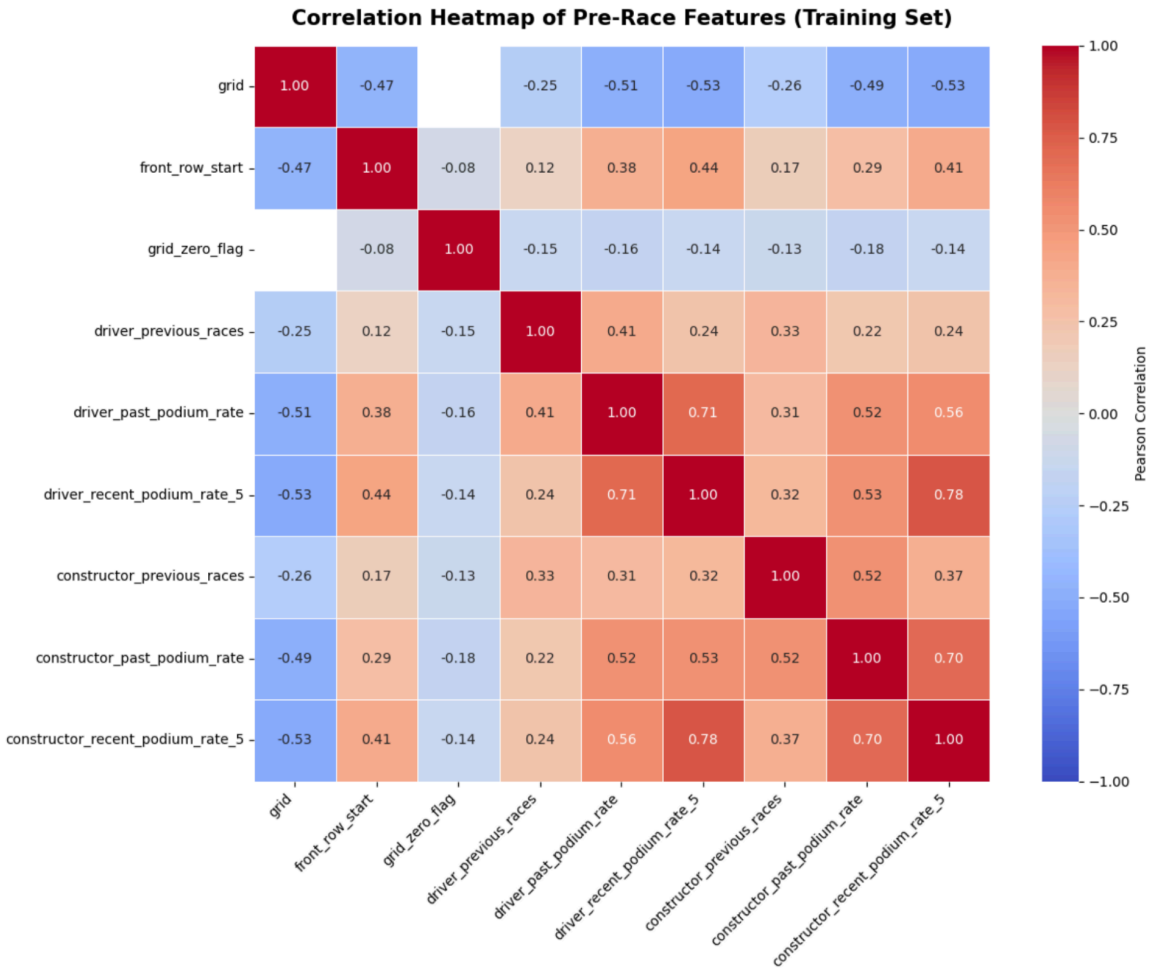


Figure 9: Pearson Correlation Heatmap of Selected Pre-Race Features (Training Set)

The heatmap illustrates the linear relationships among the nine selected predictors using training observations only. Strong positive correlations are observed between recent driver and constructor podium performance (r = 0.781), between driver historical and recent podium performance (r = 0.706), and between constructor historical and recent podium performance (r = 0.695).

Starting grid position exhibits negative relationships with several historical performance features, indicating that drivers and constructors with stronger historical results tend to start closer to the front. The heatmap also reveals overlapping information among some performance indicators. However, these features were retained for the initial experiments because they capture distinct historical and recent performance measures, and their predictive contributions were subsequently investigated through feature-group ablation.

7. Data Preprocessing and Dataset Splitting

7.1 Chronological Splitting

A chronological split is used instead of randomly distributing observations across the dataset.

This approach more closely reflects the real-world task of training on historical races and predicting future races.

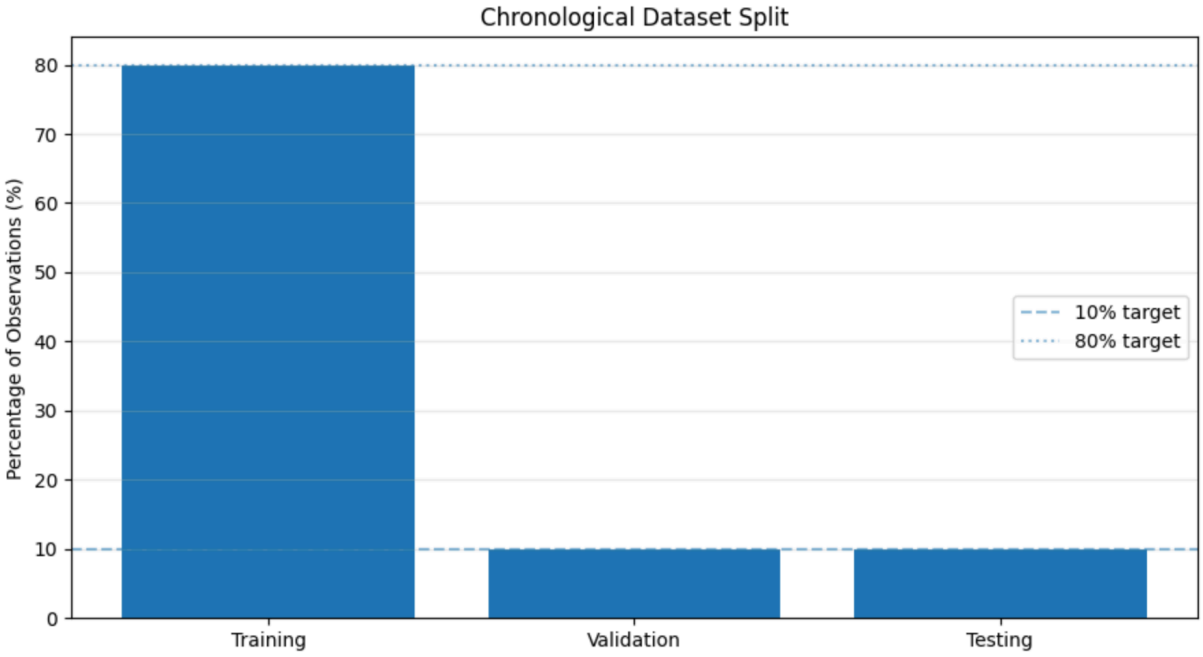
All driver observations belonging to the same race remain in the same partition.

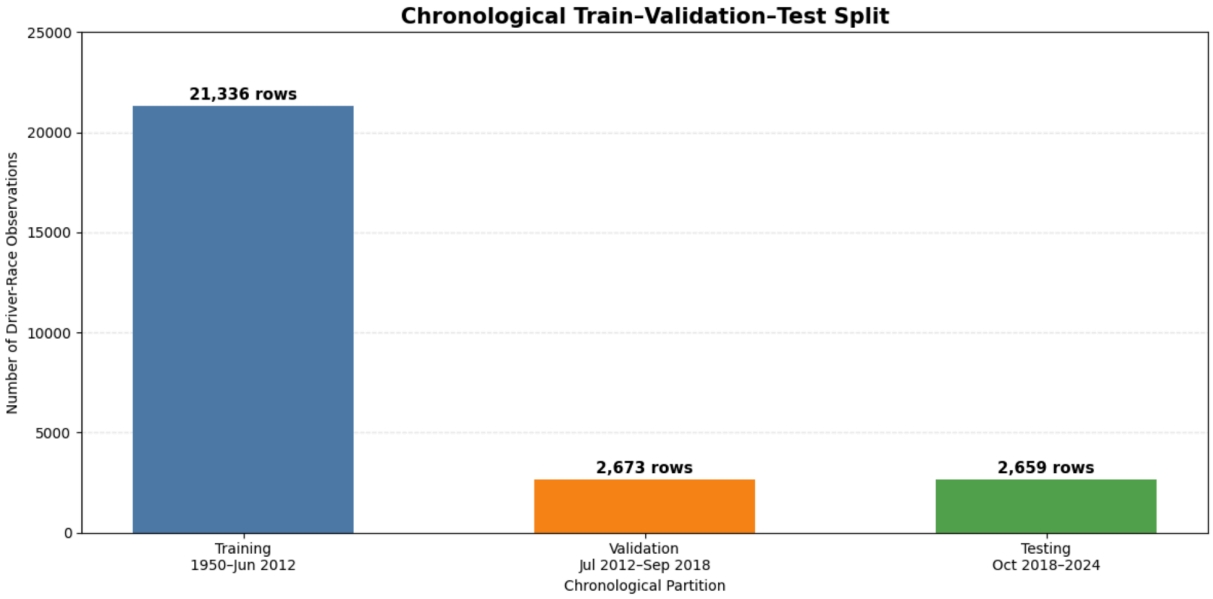
Partition	Observations	Podium rate
Training	21,336	12.27%
Validation	2,673	14.14%
Test	2,659	15.01%

The training partition covers races from May 1950 through June 2012. Validation covers July 2012 through September 2018, and testing covers October 2018 through December 2024.

The training set is used to fit model parameters, the validation set is used for model selection and threshold optimization, and the test set is reserved for final evaluation.

	Partition	Observations	Percentage	Races	First Year	Last Year	First Date	Last Date	Podium Rate (%)
0	Training	21336	80.01	866	1950	2012	1950-05-13	2012-06-24	12.27
1	Validation	2673	10.02	126	2012	2018	2012-07-08	2018-09-30	14.14
2	Testing	2659	9.97	133	2018	2024	2018-10-07	2024-12-08	15.01





	Partition	Observations	Podium Rate (%)	Period
0	Training	21336	12.27	1950–Jun 2012
1	Validation	2673	14.14	Jul 2012–Sep 2018
2	Testing	2659	15.01	Oct 2018–2024

Figure 10: Chronological Distribution of Training, Validation, and Test Observations

The chronological partitioning assigns 21,336 observations (80.0%) to training, 2,673 observations (10.0%) to validation, and 2,659 observations (10.0%) to testing.

The training partition contains the earliest races, while validation and testing contain progressively later races. This preserves temporal ordering and prevents observations from the same race from being distributed across different partitions.

Podium prevalence increases from 12.27% in training to 14.14% in validation and 15.01% in testing. These differences indicate that the class distribution varies across historical periods and should be considered when comparing model performance.

7.2 Preprocessing

Missing numerical values are handled through median imputation fitted on the training data.

Continuous numerical predictors are standardized for algorithms that benefit from feature scaling, such as Logistic Regression and the neural network.

Random Forest does not require standardization because tree-based splits are generally insensitive to linear rescaling.

All learned preprocessing parameters are fitted using training data only.

7.3 Preprocessing-Order Experiments

To investigate whether the order of preprocessing operations affects the resulting feature values and distributions, two controlled experiments were conducted using the training partition containing 21,336 observations and nine features.

A reproducible random seed of 42 was used to introduce approximately 5% additional missing values into a working copy of the training features. The same missing-value pattern was used across both experiments to ensure a controlled comparison. The original training data and finalized predictive models were not modified.

All imputation and scaling parameters were fitted using the experimental training data only. The experiments examined changes in feature distributions, minimum and maximum values, means, standard deviations, and differences between transformed observations.

Experiment 1: Median Imputation and Standardization

The first experiment compared two preprocessing sequences:

- **Configuration A: Median Imputation → StandardScaler**
- **Configuration B: StandardScaler → Median Imputation**

Metric	Configuration A	Configuration B
Minimum value	−1.6958	−1.5937
Maximum value	7.0927	6.9001
Mean	0.0000	−0.0183
Standard deviation	1.0000	0.9748
Remaining missing values	0	0

The mean absolute difference between the transformed feature values was 0.024623, while the maximum absolute difference was 0.208668.



Figure 11: Effect of Median Imputation and Standardization Order on Starting Grid Position

The histogram demonstrates that the two preprocessing orders produce similar but not identical distributions. When imputation is performed first, StandardScaler estimates its parameters using the completed dataset, resulting in an overall transformed mean close to zero and standard deviation close to one.

When standardization is performed before imputation, scaling parameters are estimated using the observed values only. Subsequently inserting standardized median values changes the final distribution, producing an overall mean of −0.0183 and standard deviation of 0.9748.

Therefore, preprocessing order affects the numerical values and distributions when median imputation is combined with standardization.

Experiment 2: Median Imputation and Min-Max Scaling

The second experiment compared:

- **Configuration A: Median Imputation → MinMaxScaler**
- **Configuration B: MinMaxScaler → Median Imputation**

Metric	Configuration A	Configuration B
Minimum value	**0.0000**	**0.0000**
Maximum value	**1.0000**	**1.0000**
Mean	**0.1410**	**0.1410**
Standard deviation	**0.2131**	**0.2131**
Remaining missing values	**0**	**0**

Both configurations produced a mean absolute difference and maximum absolute difference of 0.000000 at the displayed precision.

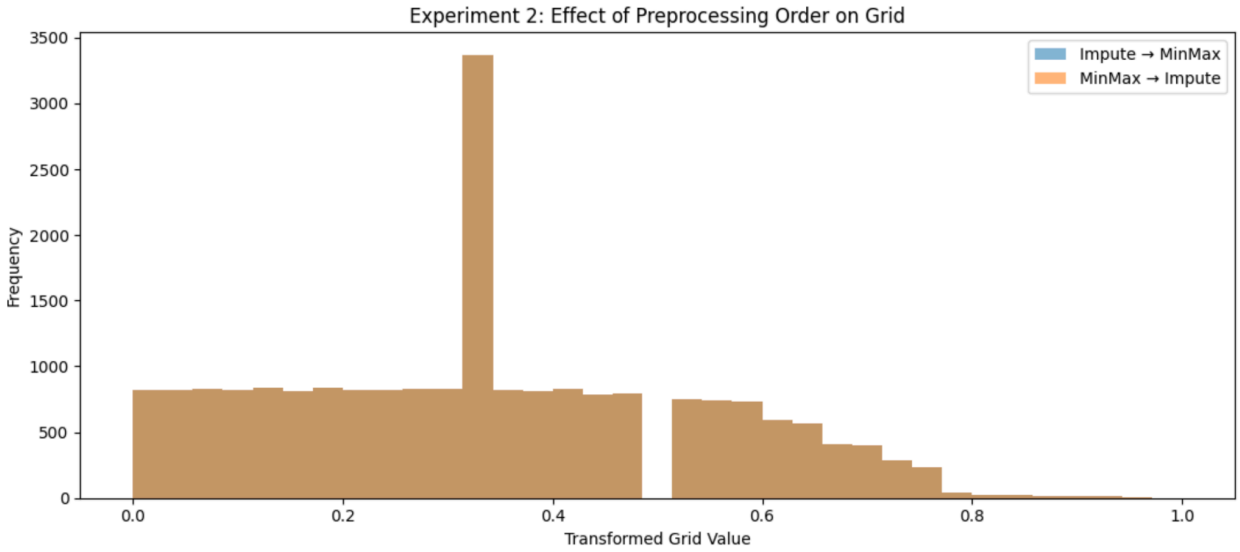


Figure 12: Effect of Median Imputation and Min-Max Scaling Order on Starting Grid Position

The histograms overlap completely, demonstrating that the two preprocessing sequences produce equivalent transformed values within numerical precision.

This occurs because MinMaxScaler performs a positive linear transformation, and median imputation is compatible with such transformations. Therefore, scaling the observed values before imputing their median produces the same transformed result as imputing the median before scaling.

Comparison and Conclusion

The experiments demonstrate that preprocessing order does not always have the same effect across different transformations.

For StandardScaler, changing the order of imputation and scaling altered the final feature distributions and numerical values. For MinMaxScaler, the two preprocessing orders produced equivalent results.

These findings support the importance of explicitly defining and validating preprocessing pipelines rather than assuming that preprocessing operations can always be reordered without changing their outputs.

For the finalized predictive models, median imputation was performed before scaling, with all preprocessing parameters learned exclusively from the training partition. This approach provides a consistent and reproducible preprocessing sequence for training, validation, testing, and inference.

8. Predictive Modeling

Three classifiers are trained using the same nine predictors and the same chronological dataset partitions.

8.1 Logistic Regression

Logistic Regression provides an interpretable statistical baseline.

The baseline model uses L2 regularization and a classification threshold of 0.50.

Final Logistic Regression Evaluation

After selecting the regularization parameter and classification threshold using training and validation data, the finalized Logistic Regression model was evaluated on the unseen chronological test partition.

The selected model used L2 regularization with C = 0.1 and an optimized classification threshold of 0.34.

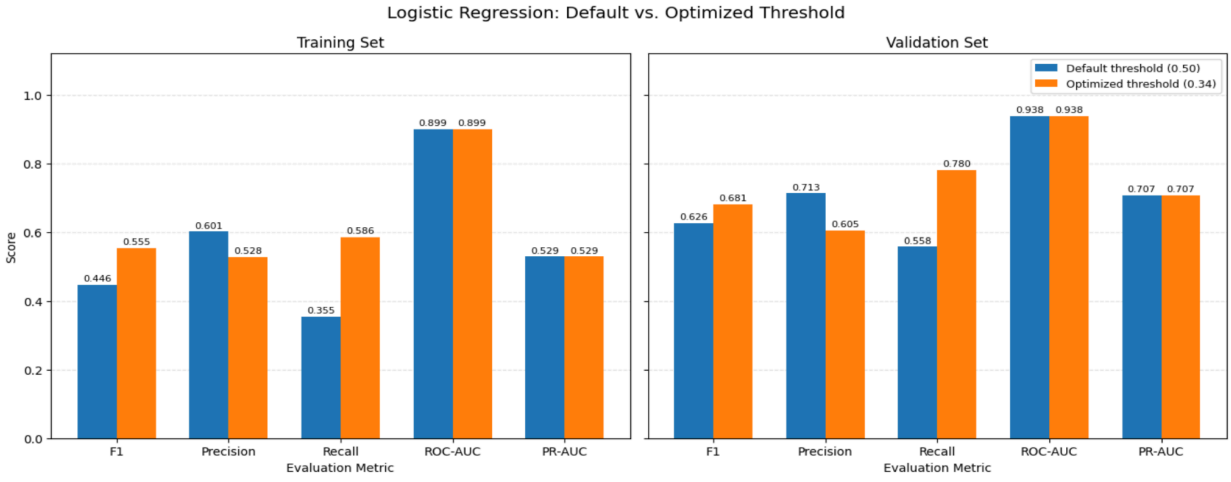
The initial Logistic Regression baseline achieved a validation F1-score of 0.6281 using the default classification threshold of 0.50. Following regularization tuning and validation-based threshold optimization, the validation F1-score increased to 0.6813 at a threshold of 0.34. This improvement demonstrates the benefit of selecting a decision threshold suited to the imbalanced podium-prediction problem. The finalized configuration was then evaluated on the held-out test partition.

	Training	Validation	Testing
Accuracy	0.8849	0.8967	0.8970
Precision	0.5279	0.6045	0.6209
Recall	0.5856	0.7804	0.8045
F1	0.5552	0.6813	0.7009
ROC-AUC	0.8987	0.9379	0.9299
PR-AUC	0.5292	0.7072	0.7050

Metric	Test score
Precision	0.6209
Recall	0.8045
F1-score	0.7009
ROC-AUC	0.9299
PR-AUC (Average Precision)	0.7050

The Logistic Regression model achieved a high recall of 0.8045, identifying approximately 80% of actual podium finishers. Its F1-score of 0.7009 indicates a favorable balance between precision and recall under the selected threshold.

The difference between precision and recall reflects the model's tendency to identify more podium candidates at the cost of additional false-positive predictions. The threshold was optimized using validation data, while the test set was used only for final evaluation.



8.2 Random Forest

Random Forest combines multiple decision trees and can model nonlinear relationships between predictors.

Random Forest: Hyperparameter Optimization and Evaluation

The Random Forest classifier was optimized using 12 hyperparameter configurations evaluated on the chronological validation partition. The selected configuration consisted of 300 decision trees, a maximum depth of 10, a minimum of two samples per leaf, and square-root feature sampling. This configuration achieved the highest validation Average Precision of 0.6923.

The classification threshold was subsequently optimized by maximizing validation F1-score, resulting in a threshold of 0.6948.

On the unseen test partition, the finalized Random Forest achieved 90.07% accuracy, 0.9258 ROC-AUC, 0.6703 Average Precision, and 0.6508 F1-score, with precision of 0.6891 and recall of 0.6165. These results indicate that the model maintains useful predictive performance on chronologically later races, although its performance is slightly lower than on the validation partition.

	Training	Validation	Testing
Accuracy	0.9042	0.8982	0.9007
Precision	0.5891	0.6152	0.6891
Recall	0.7257	0.7487	0.6165
F1	0.6504	0.6754	0.6508
ROC-AUC	0.9483	0.9371	0.9258
PR-AUC	0.7108	0.6923	0.6703

8.3 Shallow Neural Network

The third classifier was a shallow Multilayer Perceptron (MLP) trained on the same nine predictors and chronological dataset partitions as Logistic Regression and Random Forest. Numerical predictors were standardized using preprocessing fitted on the training partition.

Hyperparameter Optimization

A grid search evaluated 24 configurations combining four hidden-layer architectures, three L2 regularization strengths, and two initial learning rates. All configurations used ReLU activation, the Adam optimizer, a maximum of 300 training iterations, and a fixed random seed of 42. Model selection was based on validation Average Precision (PR-AUC).

The best configuration consisted of:

Hyperparameter	Selected value
Hidden layers	One hidden layer
Hidden neurons	16
Activation	ReLU
Optimizer	Adam
L2 regularization (alpha)	0.001

Initial learning rate	**0.005**
Training iterations	**44**
Validation PR-AUC	**0.6902**
Validation ROC-AUC	**0.9434**

The selected model completed training in 44 iterations, below the maximum of 300. Larger two-hidden-layer architectures generally produced lower validation Average Precision despite stronger training performance in some configurations, suggesting reduced generalization.

Classification Threshold Optimization

After hyperparameter selection, the classification threshold was optimized using validation F1-score. The selected threshold was 0.36, rather than the conventional threshold of 0.50.

Final Model Evaluation

Metric	Training	Validation	Test
Accuracy	**0.8858**	**0.9039**	**0.9052**
Precision	**0.5313**	**0.6410**	**0.6870**
Recall	**0.5901**	**0.7275**	**0.6767**
F1-score	**0.5592**	**0.6815**	**0.6818**

ROC-AUC	**0.9037**	**0.9434**	**0.9335**
PR-AUC	**0.5522**	**0.6902**	**0.6922**

The final MLP achieved a test ROC-AUC of 0.9335, PR-AUC of 0.6922, and F1-score of 0.6818. Its validation and test F1-scores were nearly identical, indicating stable classification performance across these two chronological partitions.

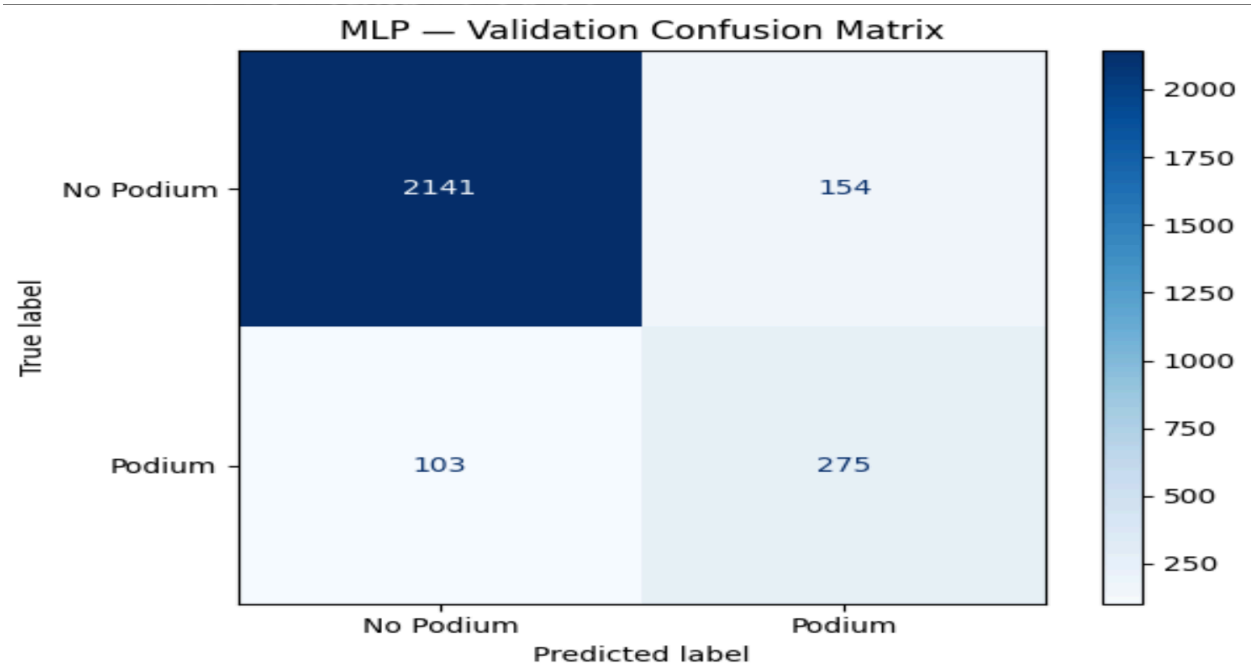
The training scores were lower than the validation and test scores. Differences in historical eras, podium prevalence, and competitive characteristics may contribute to this pattern, although the observed results alone do not establish its cause.

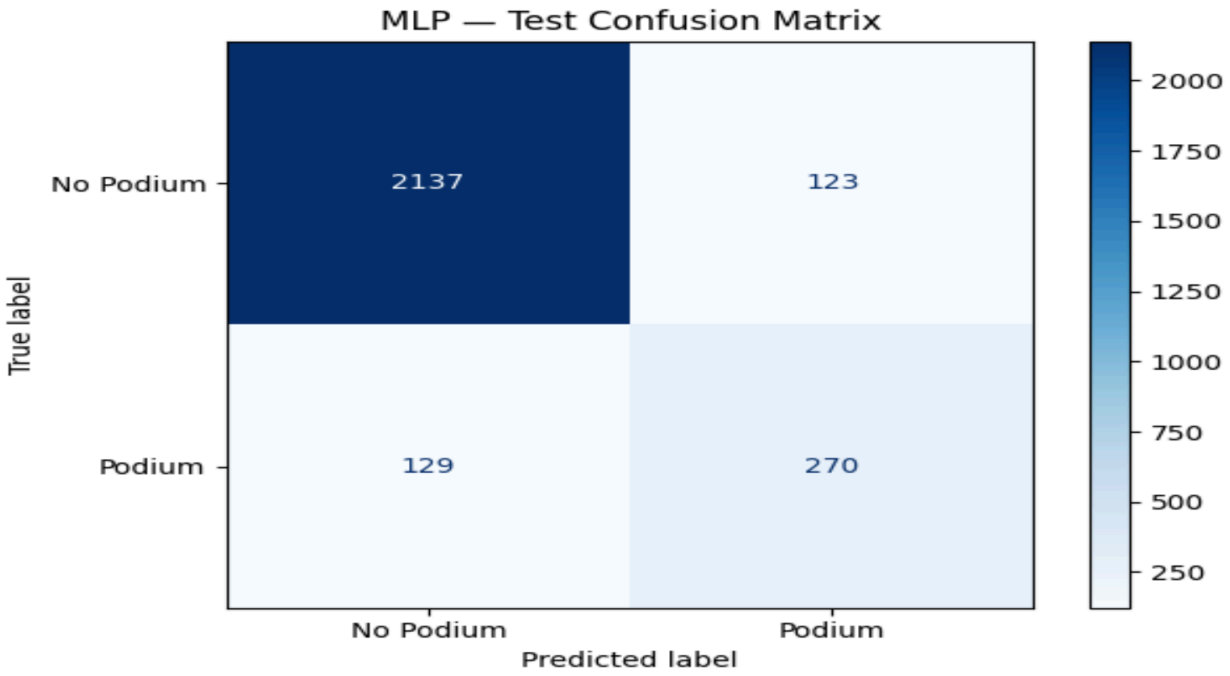
Confusion Matrix Analysis

At the selected threshold of 0.36, the validation confusion matrix contained 2,141 true negatives, 154 false positives, 103 false negatives, and 275 true positives.

The test confusion matrix contained 2,137 true negatives, 123 false positives, 129 false negatives, and 270 true positives.

Therefore, the MLP correctly identified 270 of the 399 actual podium finishes in the test partition. Compared with validation, test precision increased while recall decreased.




MLP — Test Confusion Matrix
No Podium
Podium
True label
2137
123
129
270
No Podium
Podium
Predicted label
2000
1750
1500
1250
1000
750
500
250

8.4 Model Comparison

The three finalized classifiers were evaluated using the same nine pre-race predictors and identical chronological training, validation, and test partitions.

Metric	Logistic Regression	Random Forest	MLP
Precision	0.6209	0.6891	0.6870
Recall	**0.8045**	0.6165	0.6767
F1-score	**0.7009**	0.6508	0.6818
ROC-AUC	0.9299	0.9258	**0.9335**
PR-AUC	**0.7050**	0.6703	0.6922

Logistic Regression achieved the highest test F1-score, recall, and Average Precision among the three models. The MLP achieved the highest ROC-AUC, while Random Forest achieved the highest precision by a small margin.

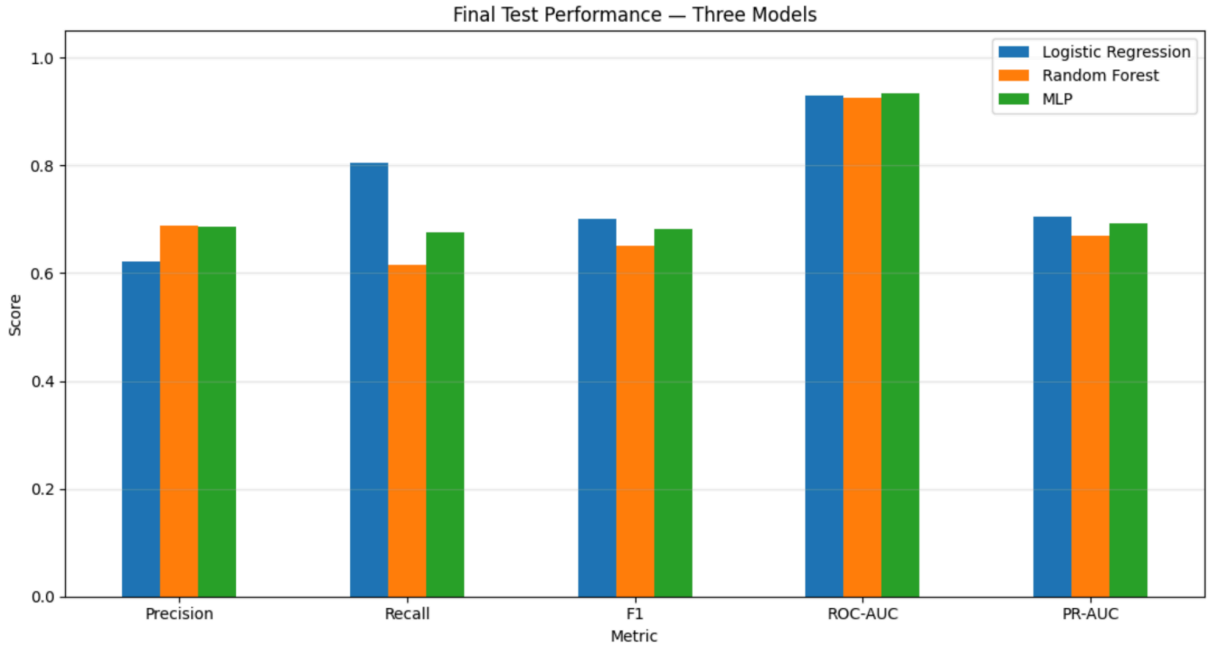
Logistic Regression's higher recall indicates that it identified a larger proportion of actual podium finishers. Its lower precision indicates that a greater proportion of its positive predictions were false positives.

The MLP provided a competitive balance of predictive performance, achieving an F1-score of 0.6818 and the highest ROC-AUC of 0.9335.

Random Forest achieved the lowest F1-score and Average Precision among the three finalized models, although it retained useful predictive performance and provided interpretable feature-importance results.

Overall, Logistic Regression is the preferred model when prioritizing F1-score, Average Precision, and podium detection recall. The MLP remains competitive when overall ranking discrimination, measured by ROC-AUC, is emphasized.

Because podium finishes are relatively rare, Average Precision and F1-score are particularly relevant metrics for this prediction task.



9. Feature-Group Ablation

Feature-group ablation investigates the contribution of starting-position features, driver-history features, and constructor-history features.

For each model, one feature group is removed at a time and the classifier is retrained.

The reduced-feature models are compared with the full-feature configuration.

Feature-group ablation was performed by independently removing starting-grid features, driver historical performance features, and constructor historical performance features. Each reduced model was retrained on the training partition and evaluated using the validation partition.

Removing grid features produced the largest performance deterioration, reducing Average Precision from 0.6923 to 0.5438 and F1-score from 0.6754 to 0.5911. Removing

driver historical performance features reduced Average Precision to 0.6491, while removing constructor historical performance features reduced it to 0.6769.

These findings indicate that starting position provides the strongest predictive contribution among the evaluated feature groups, followed by driver history and constructor history. All three groups contributed positively to validation Average Precision.

MLP — Feature-Group Ablation

Feature-group ablation was performed by independently removing grid-position features, driver historical performance features, and constructor historical performance features. Each reduced MLP was retrained, and results were averaged across three random seeds (42, 1, and 2). Classification thresholds were selected using validation F1-score.

Removed feature group	Validation ROC-AUC	Validation PR-AUC	Validation F1
None (all nine features)	0.9418	0.6804	0.6831
Grid features	0.9144	0.5992	0.6182
Driver history	0.9395	0.6686	0.6872
Constructor history	0.9426	0.7067	0.6812

Removing grid features caused the largest deterioration, reducing validation PR-AUC by 0.0811 and F1-score by 0.0649. This confirms that starting-grid information is the most important feature group for the MLP.

Removing driver-history features reduced PR-AUC by 0.0117 but slightly increased F1-score. Removing constructor-history features increased PR-AUC by 0.0263 while slightly decreasing F1-score.

These results suggest that some historical performance features may be redundant or introduce variability for the MLP. However, the improvements observed after removing a feature group do not establish that those features are universally harmful.

The ablation results are three-seed averages and therefore differ from the final single-seed MLP evaluation.

Comparison with Random Forest

Both Random Forest and MLP experienced their largest performance deterioration when grid features were removed. Random Forest's validation PR-AUC decreased by 0.1485, compared with a decrease of 0.0811 for the MLP.

However, Random Forest benefited from all three feature groups in terms of Average Precision, whereas the MLP achieved a higher mean Average Precision when constructor-history features were excluded.

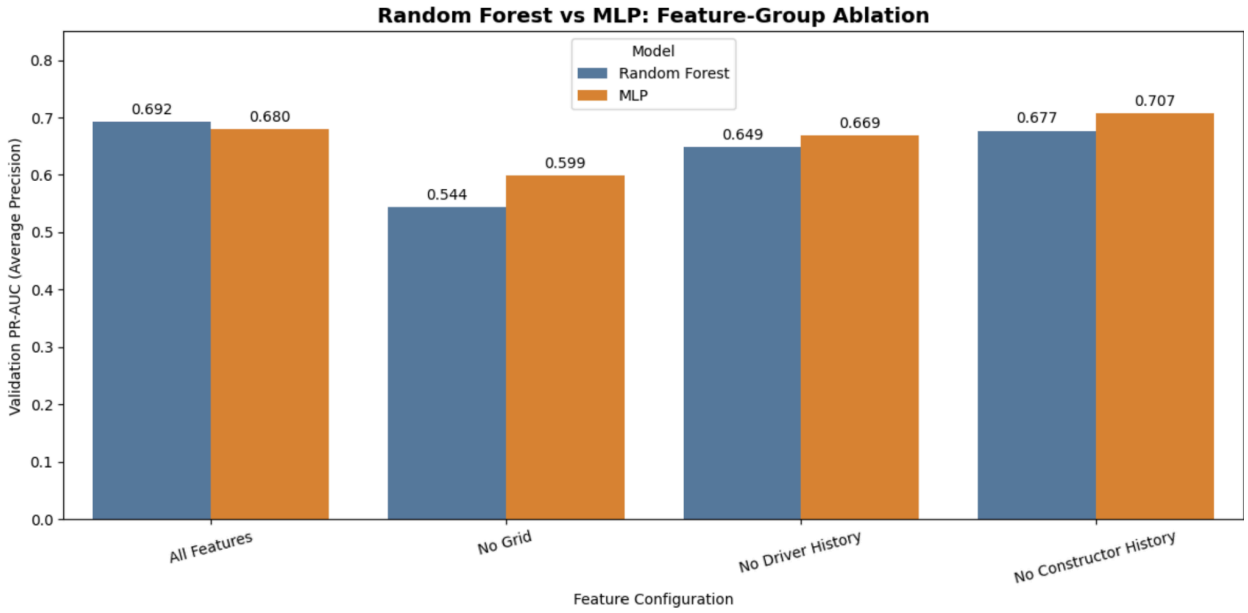


Figure 13: Random Forest and MLP Feature-Group Ablation Comparison Using Validation PR-AUC.

Logistic Regression — Feature-Group Ablation

Logistic Regression feature-group ablation evaluated four configurations: the full nine-feature model, a model without starting-position features, a model without driver-history features, and a model without constructor-history features.

Each configuration was independently trained. Regularization strength was selected using validation Average Precision, and the classification threshold was selected using validation F1-score.

In the recorded test-set analysis, the full configuration achieved an F1-score of 0.7009 and Average Precision of 0.7050. Removing starting-position features produced the largest deterioration, reducing F1-score to 0.6019 and Average Precision to 0.5764.

Removing driver-history features resulted in an F1-score of 0.6925 and Average Precision of 0.6940. Removing constructor-history features resulted in an F1-score of 0.6973, while Average Precision increased slightly to 0.7078.

These results reinforce the importance of starting-grid information. Driver history provides additional predictive information, while constructor-history contributions are smaller and depend on the evaluation metric.

The reported test comparisons are descriptive. Validation-based ablation results should be used for development decisions, and the test set should not be used to choose a preferred feature configuration.



10. Explainable Artificial Intelligence

Explainability methods are used to understand how each trained classifier uses the available predictors.

10.1 Permutation Importance

Permutation importance estimates feature relevance by measuring the reduction in predictive performance after randomly shuffling individual features.

Random Forest — Permutation Importance

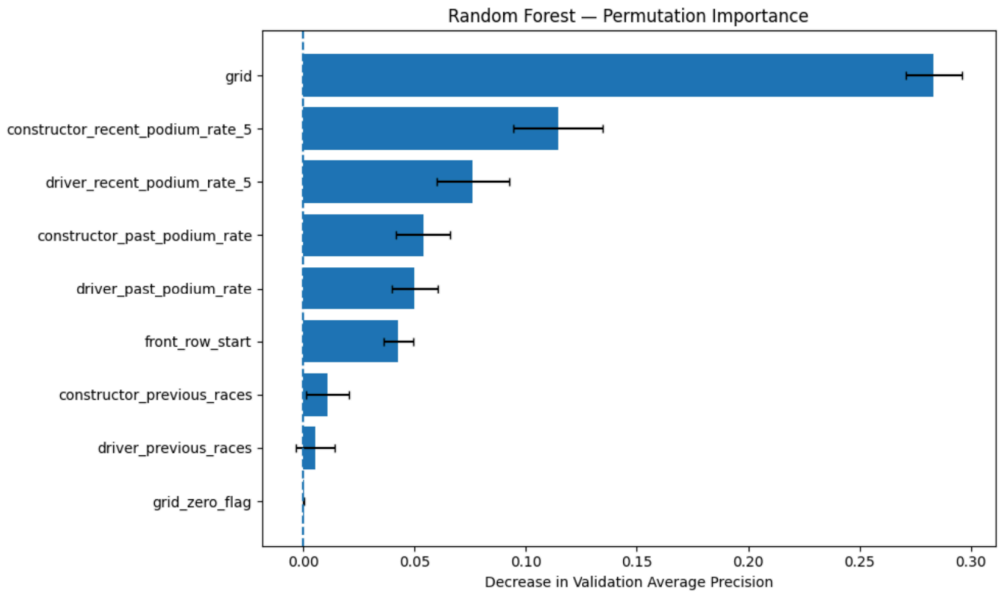
Permutation importance was calculated on the validation partition using Average Precision as the evaluation metric and 10 random permutations per feature.

Starting grid position (`grid`) was the most influential feature, producing a mean decrease in Average Precision of 0.2834 when shuffled. The next most important features were constructor recent podium rate (`constructor_recent_podium_rate_5`, 0.1145) and driver recent podium rate (`driver_recent_podium_rate_5`, 0.0762).

Historical constructor podium rate (0.0538), historical driver podium rate (0.0499), and front-row starting position (0.0427) also contributed to predictive performance.

The lowest permutation importance was observed for `grid_zero_flag` (0.0002).

These findings indicate that starting position is the strongest individual predictor for the Random Forest, while recent driver and constructor performance provide additional predictive information.



MLP — Permutation Importance

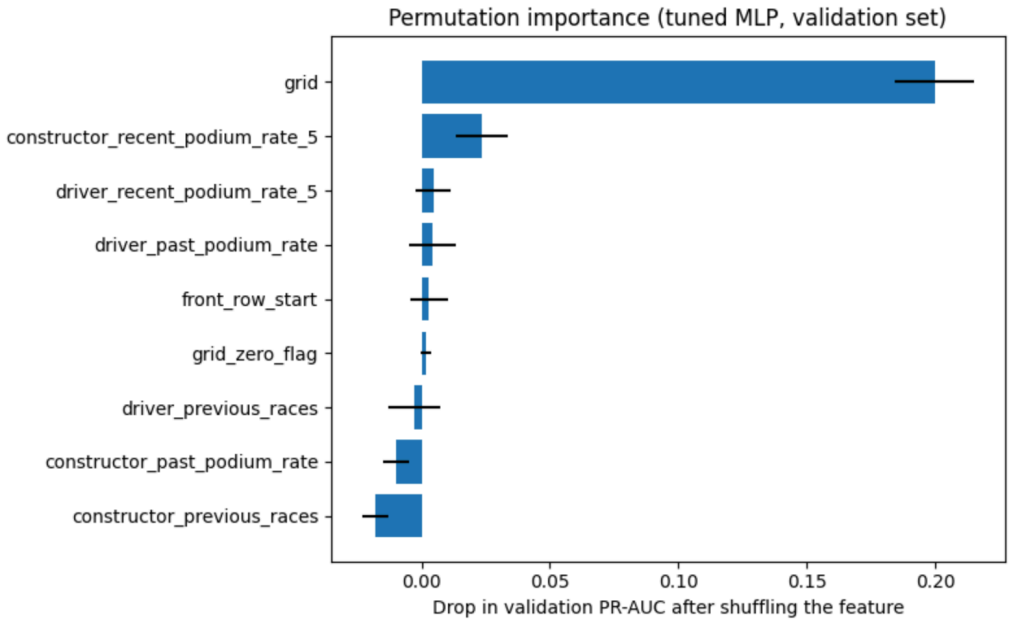
Permutation importance was evaluated on the validation partition using Average Precision as the scoring metric.

Feature	Mean importance	Standard deviation
grid	0.1997	0.0156
constructor_recent_podium_rate_5	0.0235	0.0099
driver_recent_podium_rate_5	0.0046	0.0067
driver_past_podium_rate	0.0042	0.0090
front_row_start	0.0030	0.0072
grid_zero_flag	0.0016	0.0021
driver_previous_races	−0.0028	0.0099
constructor_past_podium_rate	−0.0099	0.0049
constructor_previous_races	−0.0182	0.0051

Starting grid position was the dominant predictor, with a mean Average Precision reduction of 0.1997 when shuffled. Constructor recent podium rate was the second most influential feature, with a mean reduction of 0.0235.

Several historical features exhibited negative permutation importance, particularly constructor previous-race count and constructor historical podium rate. Negative permutation importance indicates that shuffling these features improved validation Average Precision during the experiment.

This does not mean that the features necessarily decrease the probability of a podium finish. Instead, it suggests that their information may not consistently improve the model's predictive performance on the validation partition.



Logistic Regression — Permutation Importance

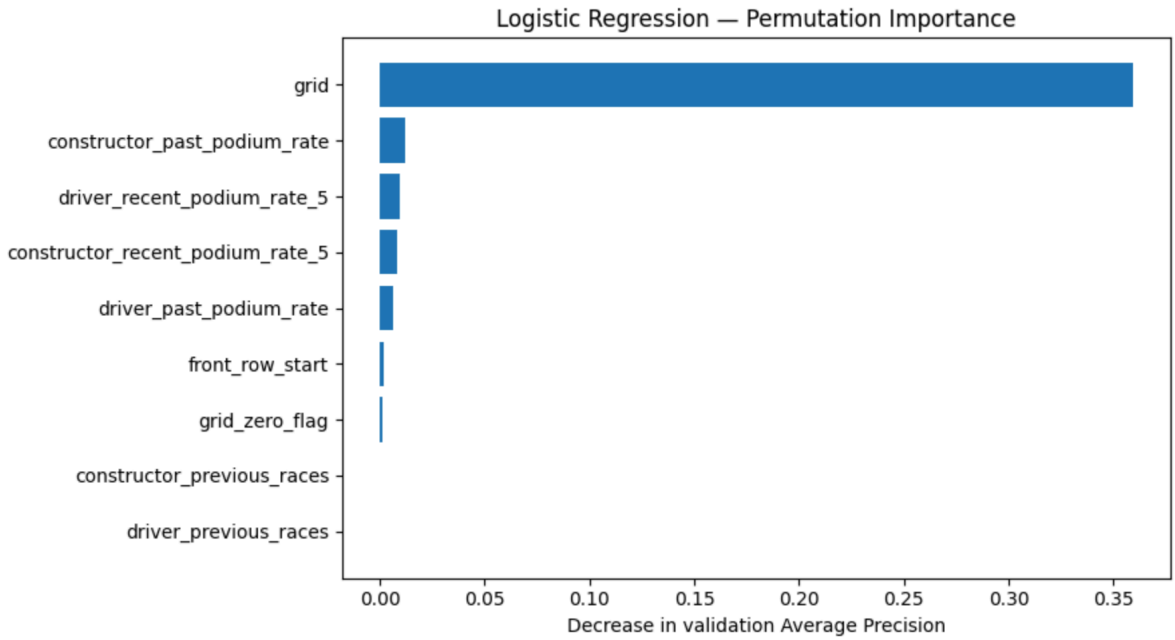
Permutation importance was calculated on the validation partition using Average Precision and ten random permutations per feature.

Starting grid position was the most influential predictor, with a mean Average Precision decrease of 0.3593 when shuffled. The next highest contributions were constructor historical podium rate (0.0123), driver recent podium rate (0.0098), constructor recent podium rate (0.0084), and driver historical podium rate (0.0061).

Front-row starting position and the grid-zero indicator had relatively small individual permutation importance values. Driver and constructor previous-race counts had importance values close to zero.

These findings suggest that the Logistic Regression model relies predominantly on starting-grid position for its predictions. Historical performance provides additional information, but its individual measured contribution is considerably smaller.

Permutation importance measures changes in predictive performance after disrupting feature values. It does not measure causal influence or the percentage contribution of a feature to an individual prediction.



10.2 SHAP

SHAP estimates individual feature contributions relative to a reference prediction.

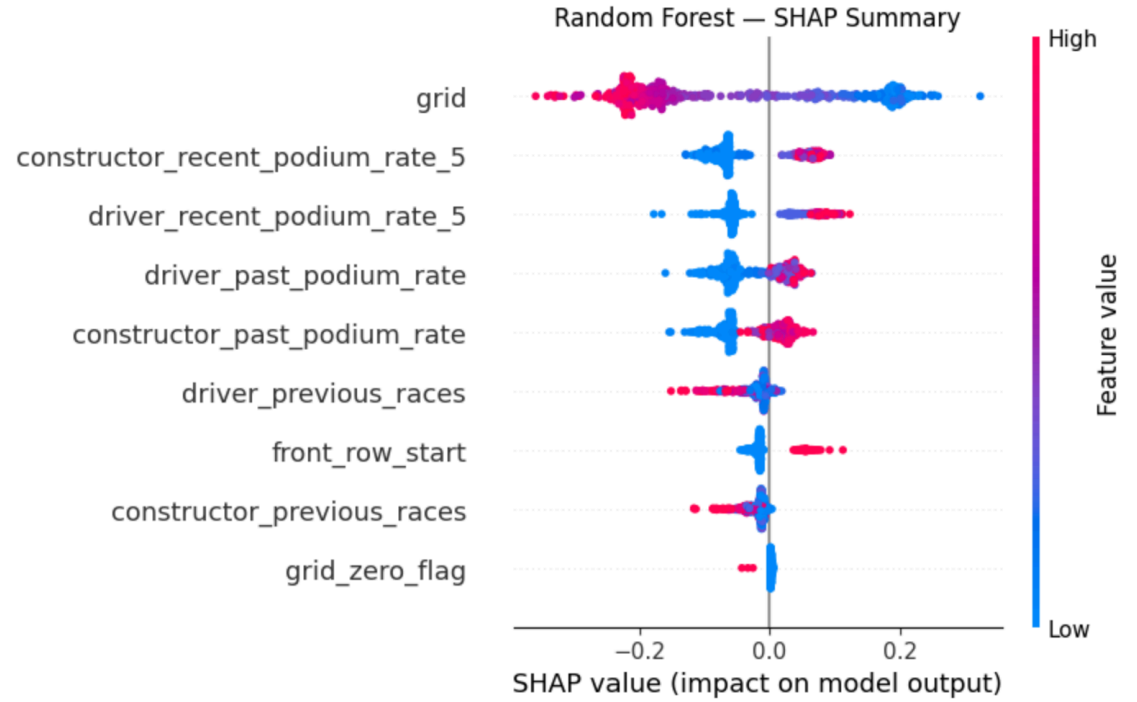
Random Forest — Global SHAP Analysis

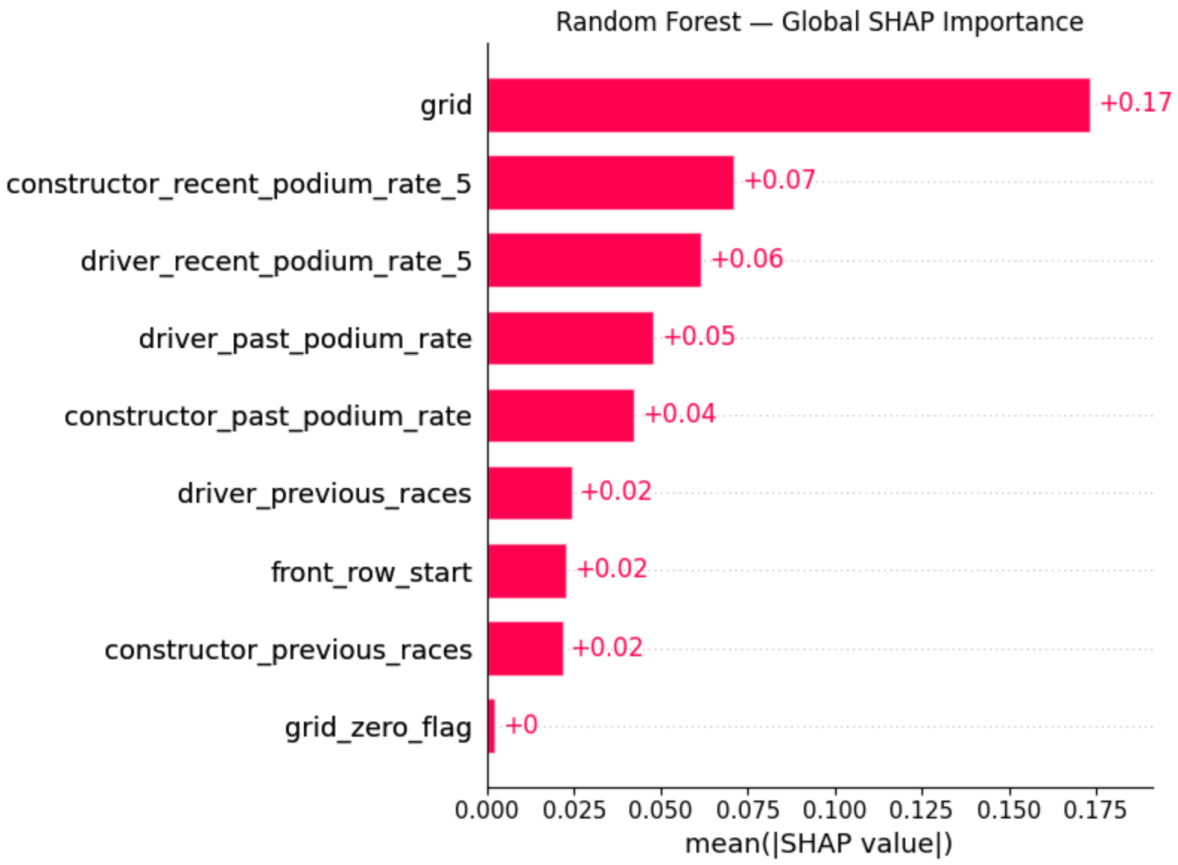
Global SHAP analysis was conducted using a sample of 500 validation observations to examine the contribution of each feature to the Random Forest's predicted podium probabilities.

The SHAP summary plot identified starting grid position as the most influential feature. Lower grid numbers, representing better starting positions, generally produced positive SHAP contributions, increasing the predicted probability of a podium finish. Conversely, higher grid numbers generally contributed negatively.

Higher recent podium rates for both constructors and drivers also tended to increase predicted podium probabilities. Historical podium rates and front-row starting position provided additional positive contributions in relevant observations.

The SHAP results are consistent with permutation importance and feature-group ablation, all of which identify starting position as the dominant predictive factor.


Random Forest — SHAP Summary
grid
constructor_recent_podium_rate_5
driver_recent_podium_rate_5
driver_past_podium_rate
constructor_past_podium_rate
driver_previous_races
front_row_start
constructor_previous_races
grid_zero_flag
−0.2
0.0
0.2
SHAP value (impact on model output)
High
Feature value
Low


Random Forest — Global SHAP Importance
grid
constructor_recent_podium_rate_5
driver_recent_podium_rate_5
driver_past_podium_rate
constructor_past_podium_rate
driver_previous_races
front_row_start
constructor_previous_races
grid_zero_flag
+0.17
+0.07
+0.06
+0.05
+0.04
+0.02
+0.02
+0.02
+0
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
mean(|SHAP value|)

MLP — Global SHAP Analysis

Global SHAP analysis was conducted using validation observations to examine the influence of the nine predictors on the MLP's podium probability estimates.

The mean absolute SHAP values identified starting grid position as the most influential feature, with an approximate contribution magnitude of 0.08. Constructor recent podium rate ranked second at approximately 0.03, followed by driver historical podium rate, driver recent podium rate, and constructor historical podium rate at approximately 0.02 each.

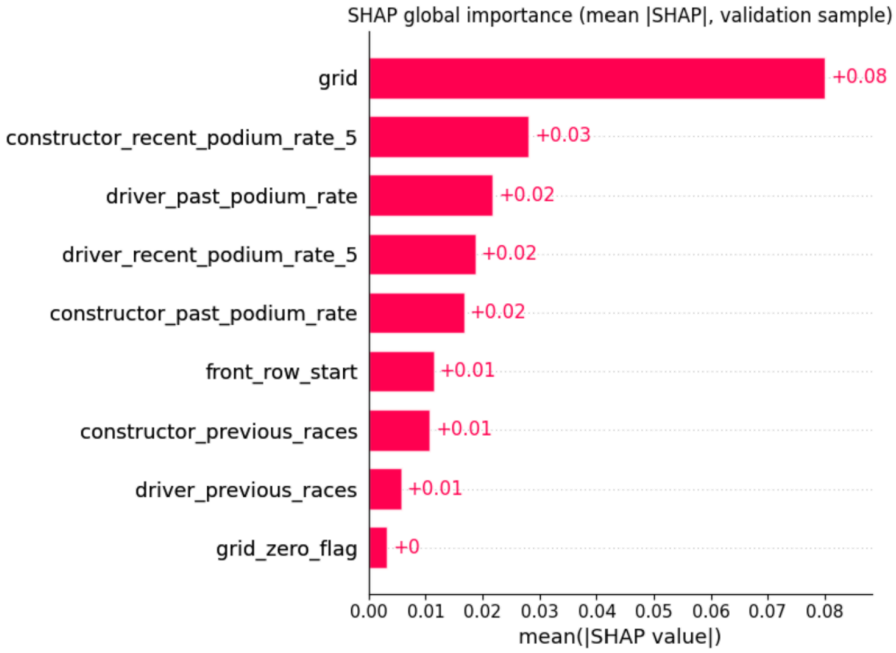
The SHAP beeswarm plot provided additional insight into the direction of these contributions.

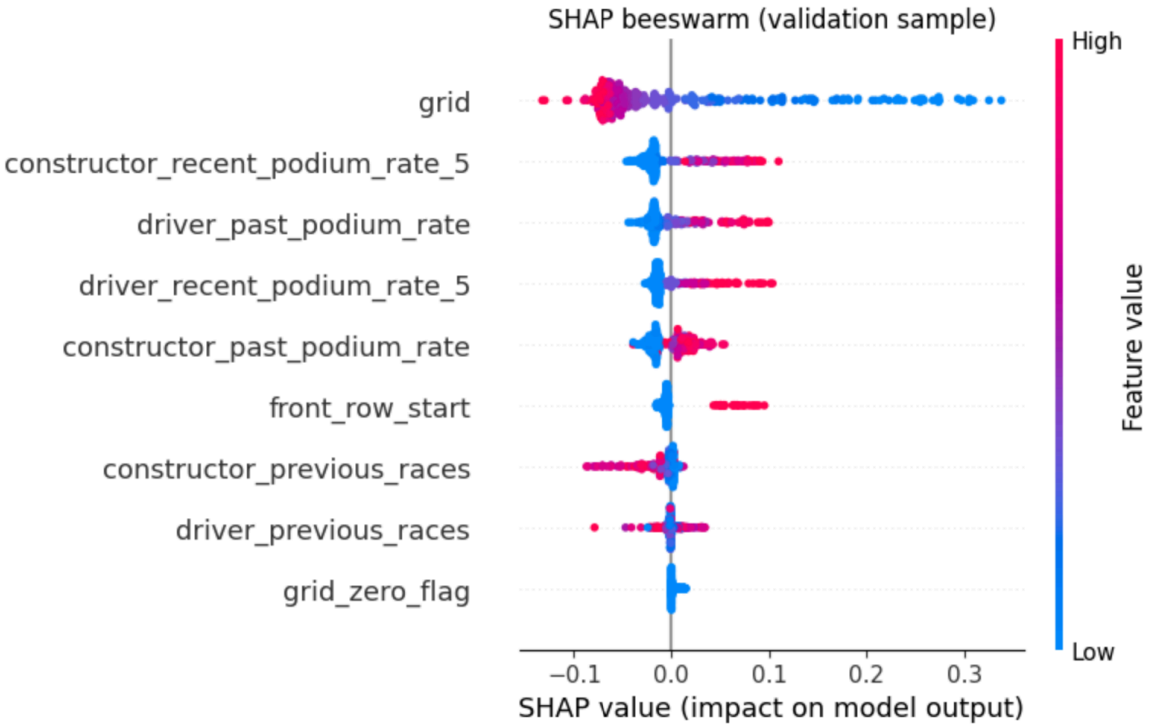
Lower starting grid positions generally produced positive SHAP values, increasing predicted podium probability. Higher grid positions tended to produce negative contributions.

Higher constructor recent podium rates, driver historical podium rates, and driver recent podium rates generally increased predicted podium probability. Conversely, higher constructor previous-race counts often contributed negatively.

The grid-zero indicator exhibited relatively small SHAP contributions.

These findings are consistent with the MLP's permutation importance and feature-group ablation results, which identify grid position as the dominant predictor.





Logistic Regression — Global SHAP Analysis

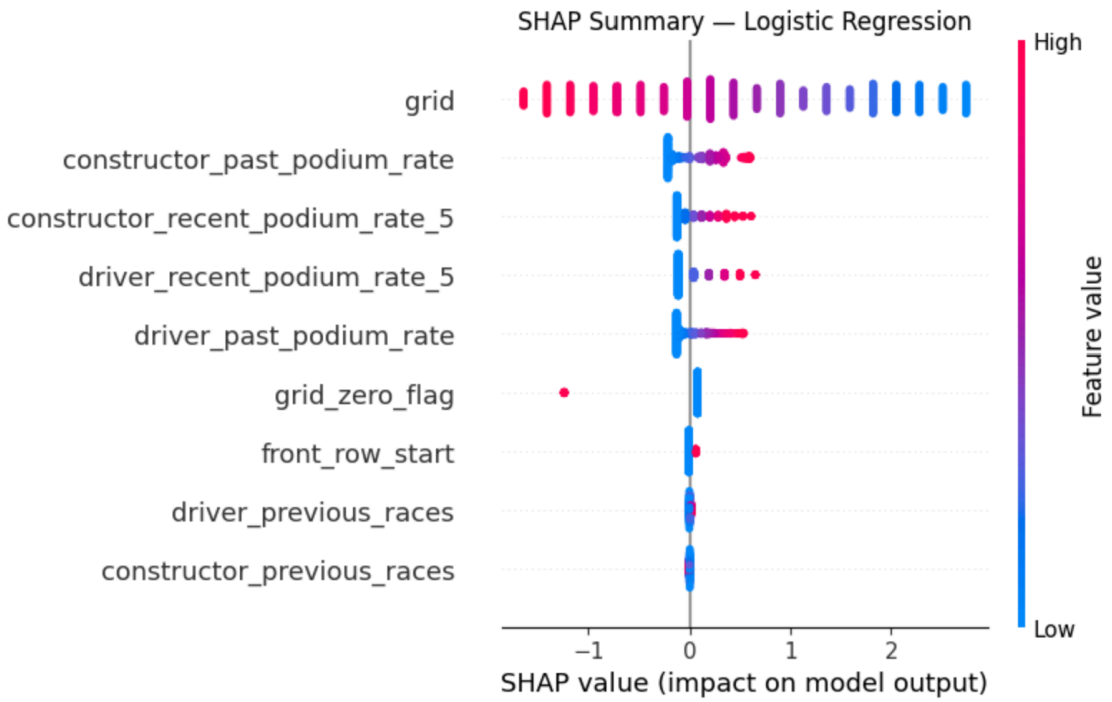
A SHAP LinearExplainer was applied to the fitted Logistic Regression classifier using 100 transformed training observations as background data and 500 transformed observations for explanation.

The explainer was verified to use the same fitted coefficients and intercept as the finalized Logistic Regression model. The nine transformed features were also verified to follow the exact order expected by the classifier.

The SHAP analysis identified starting grid position as the dominant contributor to model predictions, with historical driver and constructor performance features providing additional contributions.

For Logistic Regression, the SHAP contributions are expressed on the model's linear log-odds scale rather than directly as probability changes. Positive contributions increase the predicted log-odds of a podium relative to the reference, while negative contributions decrease them.

These explanations describe the model's learned associations and do not establish causal relationships.



10.3 Local Explanations

Local explanations investigate why a trained model produces a particular prediction. SHAP and LIME can be used to identify features that increase or decrease a prediction relative to a reference.

Random Forest — Local SHAP Waterfall Explanation

A local SHAP waterfall plot was generated for an individual observation from the validation partition.

The Random Forest predicted a **0.935 (93.5%) probability of a podium finish**, exceeding the validation-optimized classification threshold of **0.6948**. The observation was therefore classified as a predicted podium finish.

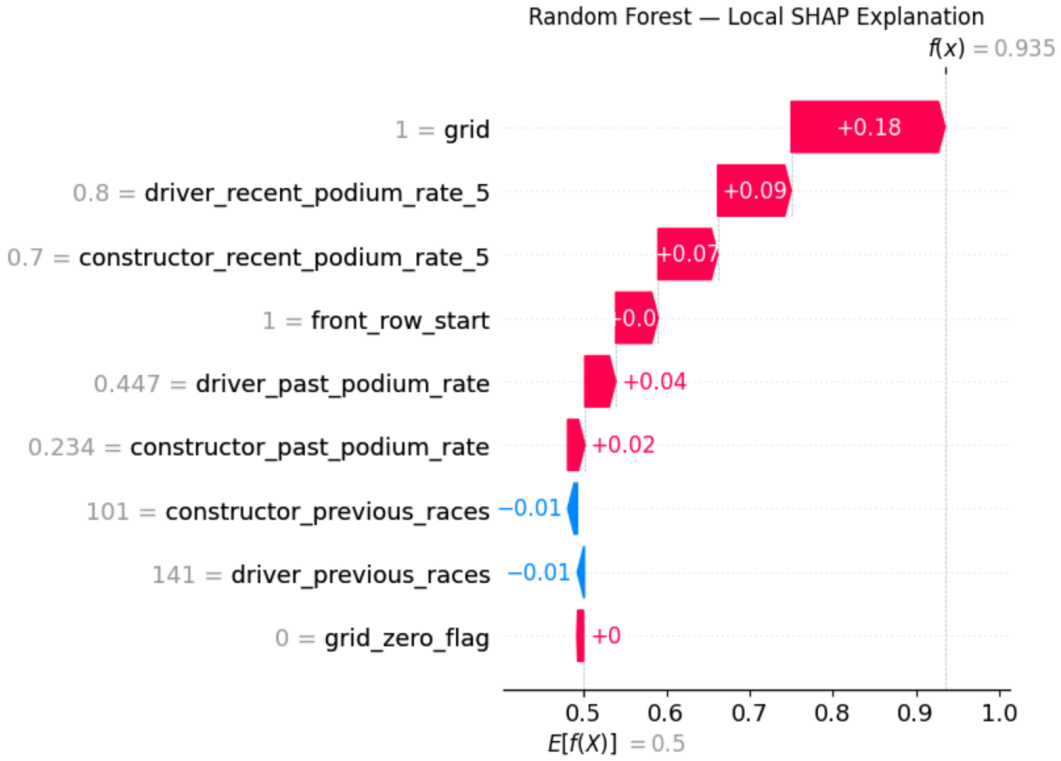
The strongest positive contribution came from starting grid position (`grid = 1`), which increased the predicted probability by approximately **0.18** relative to the SHAP baseline.

The driver's recent podium rate of **0.80** contributed approximately **+0.09**, while the constructor's recent podium rate of **0.70** contributed approximately **+0.07**.

The driver's historical podium rate (**0.447**) and constructor's historical podium rate (**0.234**) also increased the prediction, contributing approximately **+0.04** and **+0.02**, respectively.

Driver and constructor previous-race counts had small negative contributions of approximately **−0.01** each.

Overall, the local explanation demonstrates that the high predicted podium probability is primarily associated with pole position and strong recent driver and constructor performance.



MLP — Local LIME and SHAP Explanations

Local explanations were generated for a validation observation representing a driver who started third on the grid and finished on the podium.

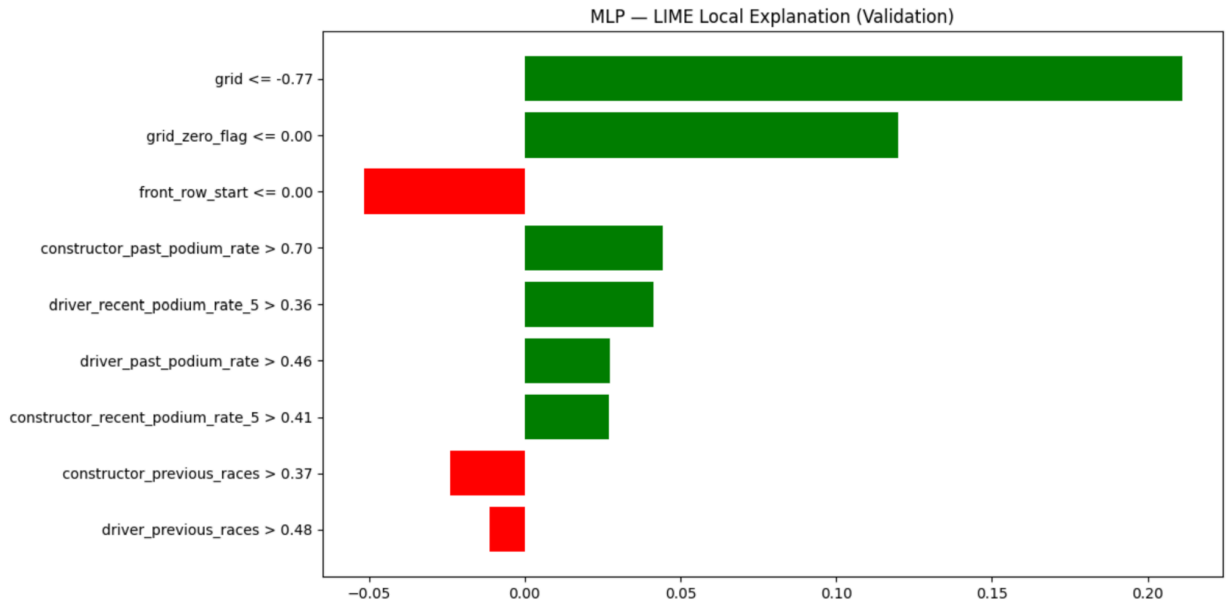
The driver's recent podium rate was 0.60, and the constructor's recent podium rate was also 0.60. The MLP assigned this observation a podium probability of **0.600**, exceeding its validation-optimized threshold of 0.36. The predicted class was therefore podium, matching the actual outcome.

LIME Explanation

LIME approximated the MLP's behavior near the selected observation using a locally fitted surrogate model.

The explanation identified favorable starting grid position as the strongest positive contributor. The absence of a front-row start contributed negatively, while recent driver performance and historical podium rates provided additional positive contributions.

The LIME plot displayed standardized or transformed feature values because the explanation was generated using the MLP's preprocessed inputs.



SHAP Waterfall Explanation

The SHAP waterfall explained the same validation observation relative to a baseline podium probability of approximately **0.134**.

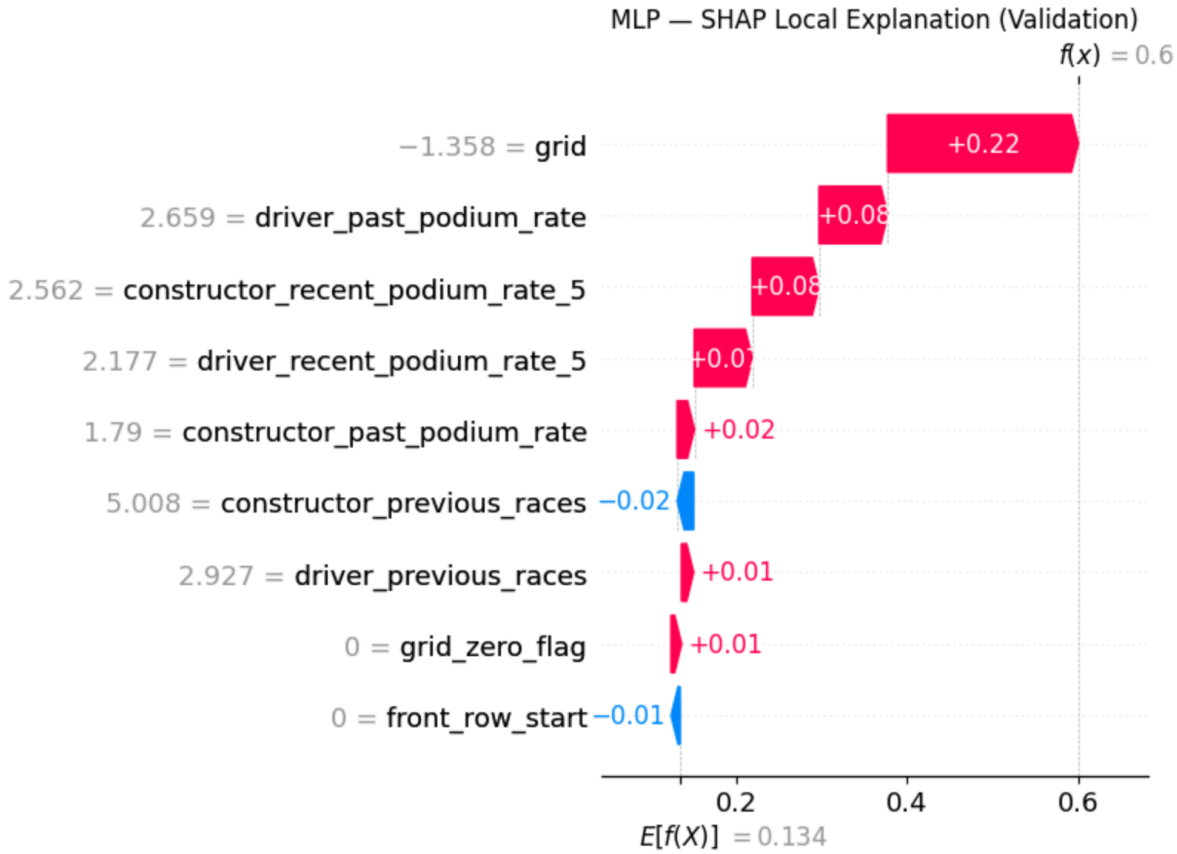
The strongest positive contribution came from grid position, contributing approximately **+0.22**. Driver historical podium rate and constructor recent podium rate each contributed approximately **+0.08**.

Driver recent podium rate and constructor historical podium rate also contributed positively. Constructor previous-race count contributed approximately **−0.02**, while the absence of a front-row start contributed approximately **−0.01**.

Together, these contributions increased the model's predicted podium probability from approximately 0.134 to 0.600.

Both LIME and SHAP highlighted favorable grid position and historical competitive performance as important reasons for the positive prediction. Their numerical contributions are not directly interchangeable because LIME uses a local surrogate approximation, whereas SHAP allocates contributions relative to a reference prediction.

driver_previous_races > 0.48: -0.0113



Logistic Regression — Local SHAP and LIME Explanations

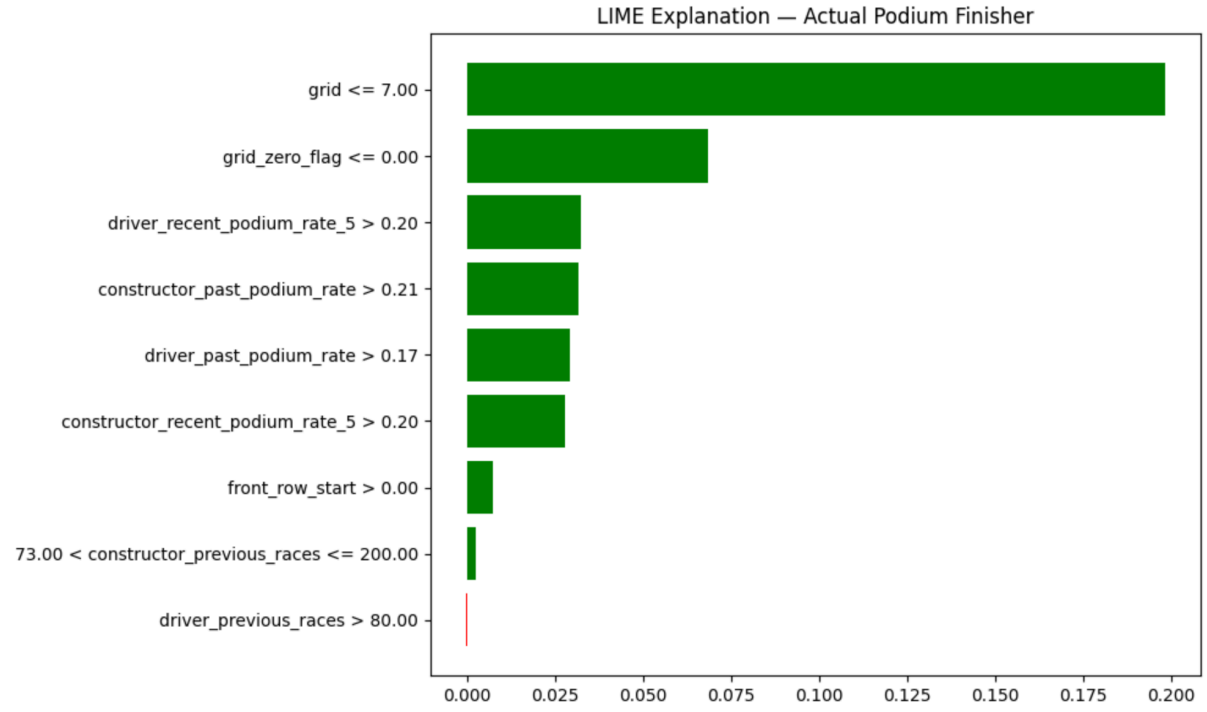
Local explanations were generated to investigate individual Logistic Regression podium predictions.

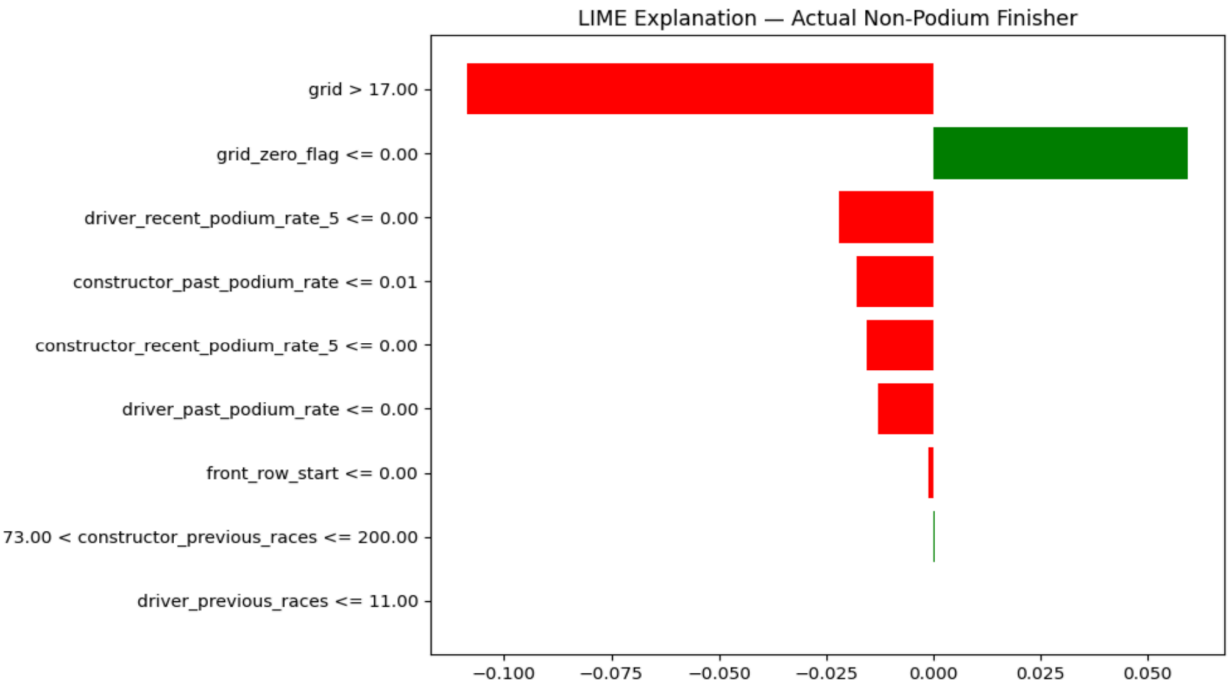
SHAP decomposes an individual prediction into feature contributions relative to a reference output. LIME approximates the classifier locally by generating perturbed observations and fitting an interpretable surrogate model around the selected example.

The Logistic Regression LIME prediction function uses the finalized model pipeline, ensuring that perturbed observations receive the same training-fitted preprocessing as ordinary inference inputs.

For local Logistic Regression explanations, one actual podium finisher and one actual non-podium finisher were selected from the test partition. LIME was used to approximate the model's prediction locally and identify features supporting or opposing the podium class. These explanations were generated using the finalized pipeline and training-fitted preprocessing.

The local plots illustrate how starting-grid position and historical performance features contribute to individual predictions. The explanation weights describe the behavior of the fitted model around each observation and should not be interpreted as causal effects.





11. Inference Function

The prediction pipeline supports inference using either a complete pre-race feature record or a partially missing record.

For the MLP, input features are arranged in the same order as during training and transformed using the previously fitted preprocessing pipeline. Missing numerical values are handled using the training-fitted imputation strategy. The model then produces a podium probability and applies the validation-selected classification threshold of 0.36.

Complete-Record Inference

A complete validation observation was passed to the inference function without missing feature values.

The model produced a podium probability of 0.1205, below the threshold of 0.36, and therefore classified the observation as non-podium.

The inference result matched the model's direct validation prediction of 0.1205, confirming consistent preprocessing and prediction behavior for this example.

Partially Missing-Record Inference

A second prediction was generated using the same observation after removing three historical features:

- **Driver historical podium rate**
- **Driver recent podium rate over five races**
- **Constructor recent podium rate over five races**

The preprocessing pipeline handled these missing values, and the MLP produced a podium probability of 0.0336, again resulting in a non-podium classification.

Inference scenario	Podium probability	Predicted class	Missing features
Complete record	0.1205	Non-podium	0

Partially missing record	**0.0336**	**Non-podium**	**3**

These examples demonstrate that the MLP can generate predictions from complete and partially missing inputs without refitting preprocessing on the inference data.

The probability difference between the two examples illustrates the model's sensitivity to the available historical information, but it should not be interpreted as a general causal effect of missing features.

Logistic Regression — Complete and Partially Missing Inference

The finalized Logistic Regression pipeline was tested using complete and partially missing pre-race feature records.

The pipeline uses training-fitted median imputation for missing numerical values and standardization for continuous predictors. The optimized classification threshold is 0.34.

Inference scenario	Podium probability	Predicted class
Complete record	**0.1922**	**Non-podium**
Partially missing record	**0.1059**	**Non-podium**

For the complete record, the inference function returned a podium probability of 0.1922, matching the direct model prediction.

For the partially missing record, three historical predictors were omitted: driver historical podium rate, driver recent podium rate, and constructor recent podium rate. The pipeline handled the missing values using its fitted imputation strategy and returned a probability of 0.1059.

Both examples were classified as non-podium because their predicted probabilities were below the selected threshold of 0.34.

These tests demonstrate that the finalized Logistic Regression pipeline supports consistent inference for complete and partially missing input records without refitting preprocessing.

```
COMPLETE RECORD
{'podium_probability': 0.1922, 'predicted_podium': False, 'threshold': 0.34, 'missing_features_imputed': []}
Direct model probability: 0.1922

PARTIALLY MISSING RECORD
{'podium_probability': 0.1059, 'predicted_podium': False, 'threshold': 0.34, 'missing_features_imputed': ['driver_past_podium_rate', 'driver_recent_podium_rate_5',
'constructor_recent_podium_rate_5']}
```

12. Limitations and Conclusion

12.1 Limitations

Several limitations should be considered when interpreting the results.

First, Formula 1 has changed considerably between 1950 and 2024. Differences in regulations, reliability, and field sizes may affect the relationships learned from historical data.

Second, detailed qualifying data is less consistently available for older seasons, limiting the use of some potentially informative qualifying predictors.

Third, constructor identities may change following team renaming or restructuring, potentially interrupting historical performance records.

Fourth, the selected predictors cannot represent every relevant factor, including unforeseen incidents, race strategy, and changing weather conditions.

Finally, predictive feature importance does not establish causality.

12.2 Conclusion

This project developed a complete data-engineering and machine-learning workflow for predicting Formula 1 podium finishes using historical race information available after qualifying and before the race.

The data-engineering stage transformed relational race records into a validated driver-race analytical dataset containing 26,668 observations. The analysis addressed front-row-to-podium conversion across circuits and eras, home-country performance after controlling for starting position, and constructor mechanical reliability across two modern regulatory periods.

Three classifiers were investigated: Logistic Regression, Random Forest, and a shallow Multilayer Perceptron. All models used the same nine predictors and chronological training, validation, and test partitions.

On the recorded unseen test results, Logistic Regression achieved the highest F1-score (0.7009), recall (0.8045), and Average Precision (0.7050). The MLP achieved the highest ROC-AUC (0.9335), while Random Forest achieved the highest precision (0.6891).

Feature-group ablation, permutation importance, and SHAP consistently identified starting-grid position as the dominant predictive factor. Historical driver and constructor performance provided additional information, although their relative contributions varied across models.

The inference experiments demonstrated that the trained pipelines could produce predictions from complete and partially missing feature records using training-fitted preprocessing.

Overall, the results demonstrate that useful podium predictions can be generated from pre-race information without relying on current-race outcomes. Logistic Regression provides the strongest overall performance under the project's F1-score and Average Precision criteria, while the MLP offers competitive predictive discrimination.

The findings remain subject to limitations arising from historical changes in Formula 1, missing older qualifying records, constructor identity changes, and unobserved race-day factors. Feature importance and explainability results describe predictive associations rather than causal effects.

References

1. DMET904, *Milestone 1: Formula 1 Podium Prediction Companion*, project specification.
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3. Kaggle, *Formula 1 World Championship (1950–2024)*, historical Ergast dataset.